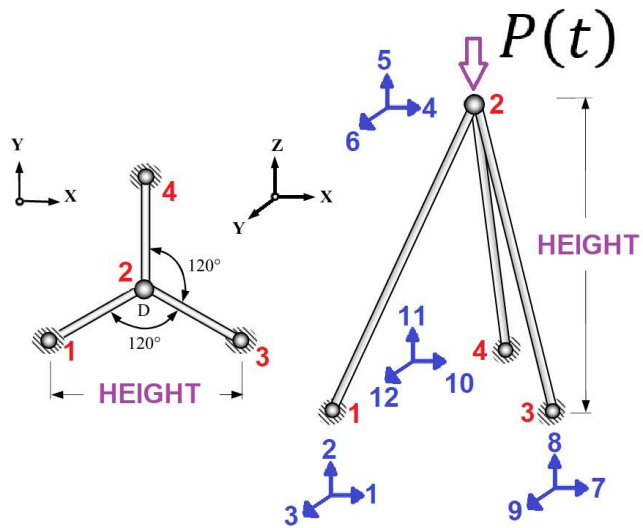


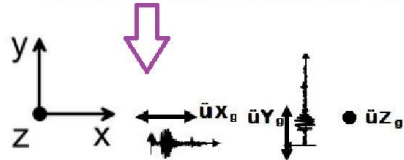
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF THREE ELEMENTS 3D TRUSS: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



INCREMENTAL DISPLACEMENT



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

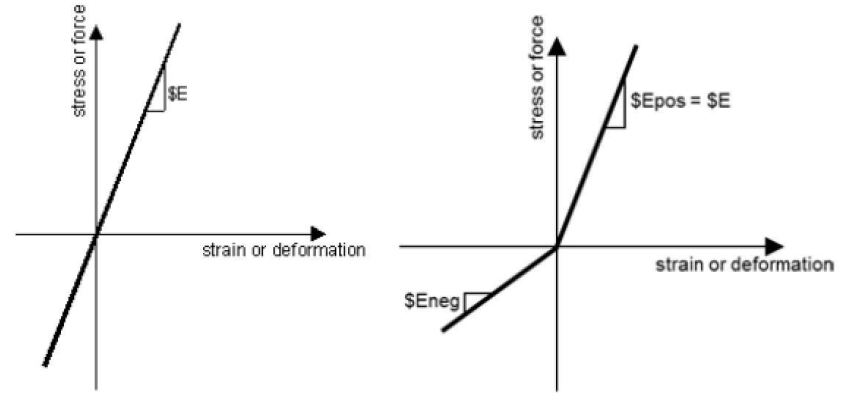
Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

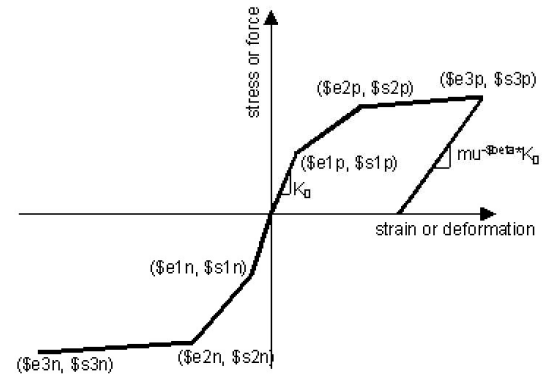
Δ_u = Lateral Ultimate Displaement from Pushover Analysis

$$P(t) = P_0 e^{-0.05 \bar{\omega} t} \sin(\bar{\omega} t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL



INELASTIC MATERIAL

Spyder (Python 3.12)

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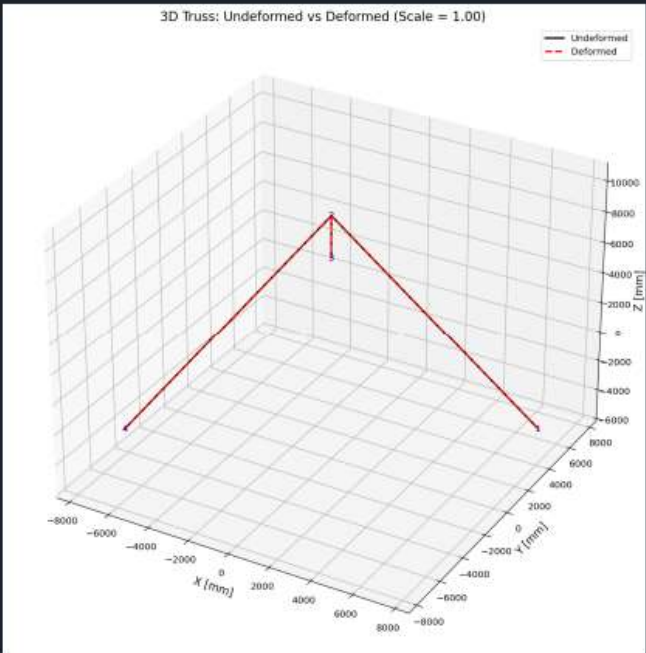
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MENTS_8_ANA\p(t)_3D_TRUSS_THREE_ELEMENTS_8_ANA.py

P(t)_3D_TRUSS_THREE_ELEMENTS_8_ANA.py

PLOT_3D_TRUSS.py

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF THREE ELEMENTS 3D TRUSS : AN OPENSEE
4 # FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-H
5 #
6 # P(t) = P0 sin(wt)
7 # P(t) = P0 exp(-0.05wt) sin(wt)
8 #
9 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND E
10 # OF ENERGY DISSIPATION CAPACITY INDEX
11 #
12 # THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
13 # EMAIL: salar.d.ghashghaei@gmail.com
14 #####
15 """
16 Nonlinear Seismic Performance Assessment of Three Elements 3D Truss:
17 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and
18
19 This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of hree Ele
20 for performance-based earthquake engineering. The 2D model incorporates both material non
21 (elastic-perfectly plastic or hysteretic steel with strain hardening)
22 and geometric nonlinearity to capture P-delta effects
23 and large displacements-essential for collapse assessment.
24
25 Eight analysis protocols are implemented:
26 (1) [PERIOD] : Structural Period
27 (2) [STATIC] : Gravity load analysis establishing dead Load state
28 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
29 and plastic mechanism identification
30 (4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturing hysteresis,
31 pinching behavior, and energy dissipation degradation
32 (5) [STATIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Static Analysis of External time-dependent
33 (6) [DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Dynamic Analysis of External time-dependen
34 (7) [FREE-VIBRATION] : Free-vibration with initial conditions extracting damping ratios
```

3D Truss: Undeformed vs Deformed (Scale = 1.00)



[Python Console] Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 34%

Spyder (Python 3.12)

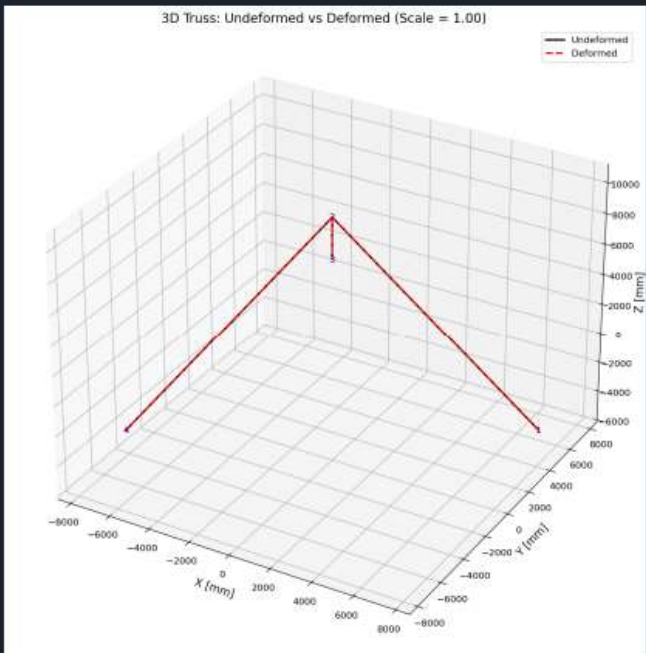
File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...34_3D_TRUSS_THREE_ELEMENTS_8_ANA\PLOT_3D_TRUSS.py

P(t)_3D_TRUSS_THREE_ELEMENTS_8_ANA.py X PLOT_3D_TRUSS.py X

```
1 def PLOT_3D_TRUSS(deformed_scale=1.0):
2     """
3     Plot 3D Truss geometry with undeformed (black) and deformed (red dashed) shapes from Ope
4
5     Parameters
6     -----
7     deformed_scale : float, optional
8         Scale factor for visualizing deformations (defaults to 1.0)
9
10    THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAIE (QASHQAI)
11    """
12    import openseespy.opensees as ops
13    import numpy as np
14    import matplotlib.pyplot as plt
15    from mpl_toolkits.mplot3d import Axes3D
16
17    # Initialize figure and 3D axes
18    fig = plt.figure(figsize=(14, 10))
19    ax = fig.add_subplot(111, projection='3d')
20
21    # Gather node coordinates
22    nodes = ops.getNodeTags()
23    node_coords = {node: ops.nodeCoord(node) for node in nodes}
24
25    # Plot undeformed shape (black solid lines)
26    for i, ele in enumerate(ops.getEleTags()):
27        node1, node2 = ops.eleNodes(ele)
28        x1, y1, z1 = node_coords[node1]
29        x2, y2, z2 = node_coords[node2]
30        ax.plot([x1, x2], [y1, y2], [z1, z2],
31                'k-', linewidth=2,
32                label='Undeformed' if i == 0 else "")
33
34    # Plot deformed shape (scaled, red dashed lines)
```

3D Truss: Undeformed vs Deformed (Scale = 1.00)



Python Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 33%

STATIC ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MENTS_8_ANA\P(t)_3D_TRUSS_THREE_ELEMENTS_8_ANA.py

P(t)_3D_TRUSS_THREE_ELEMENTS_8_ANA.py x PLOT_3D_TRUSS.py x

```
263
264
265 if ANAL_TYPE == 'STATIC':
266     ops.integrator('LoadControl', 1.0)
267     ops.analysis('Static')
268     OK = ops.analyze(1)
269     S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
270     ops.reactions()
271     reaction.append(ops.nodeReaction(1, 3)+ops.nodeReaction(3, 3)+ops.nodeReaction(4,
272     disp_mid.append(ops.nodeDisp(center_node, 3)) # DISPLACEMENT NOD
273     dispX.append(ops.nodeDisp(center_node, 1)) # DISPLACEMENT NOD
274     dispY.append(ops.nodeDisp(center_node, 2)) # DISPLACEMENT NOD
275     dispZ.append(ops.nodeDisp(center_node, 3)) # DISPLACEMENT NOD
276     # Store axial forces, strain and stress
277     for ele_id in ele_axialforce.keys():
278         ele_axialforce[ele_id].append(ops.eleResponse(ele_id, 'axialForce')) # [N]
279         ele_stress[ele_id].append(ops.eleResponse(ele_id, 'material', 'stress')) # [N/
280         ele_strain[ele_id].append(ops.eleResponse(ele_id, 'material', 'strain')) # [mm
281         if ELE_TYPE == 'INELASTIC':
282             ele_di[ele_id].append(100*(abs(ele_strain[ele_id]) - ey) - (esu - ey))
283         if ELE_TYPE == 'ELASTIC':
284             ele_di[ele_id].append(0.0)
285     # Store displacements
286     for node_id in node_displacementsX.keys():
287         node_displacementsX[node_id].append(ops.nodeDisp(node_id, 1))
288         node_displacementsY[node_id].append(ops.nodeDisp(node_id, 2))
289         node_displacementsZ[node_id].append(ops.nodeDisp(node_id, 3))
290     else:
291         print('\n\nSTATIC ANALYSIS DONE.\n\n')
292
293 DATA = (reaction, disp_mid,
294         ele_axialforce, ele_stress, ele_strain, ele_di,
295         node_displacementsX, node_displacementsY,
296         dispX, dispY)
```

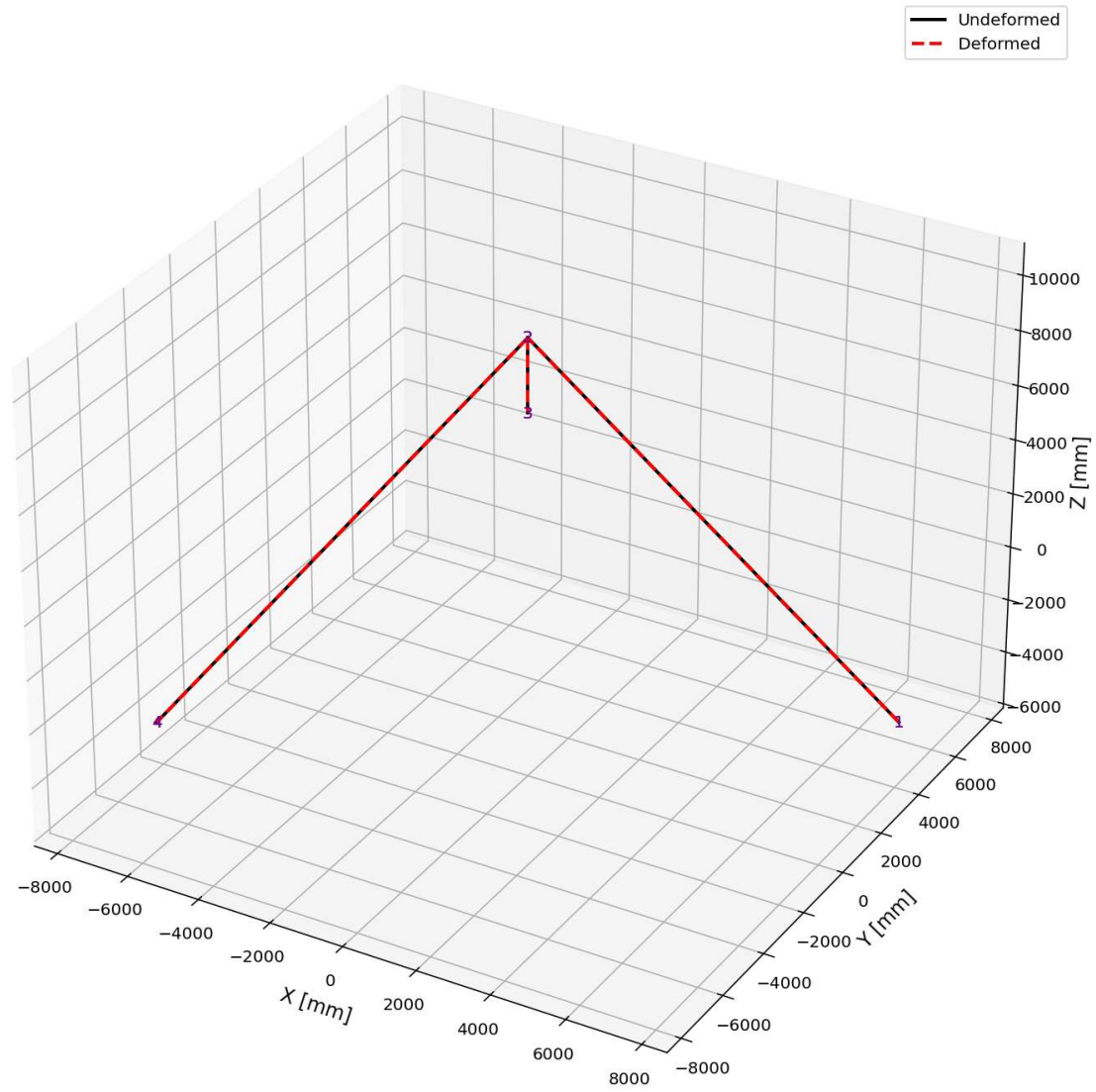
3D Truss: Undeformed vs Deformed (Scale = 1.00)

Legend: Undeformed (black line), Deformed (red line)

Python Console | Files | Help | Variable Explorer | Debugger | Plots | History

Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 1, Col 1 | UTF-8 | CRLF | RW | Mem 33%

3D Truss: Undeformed vs Deformed (Scale = 1.00)



PUSHOVER ANALYSIS

Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\EIGHT_ANALY...MENTS_8_ANA\P(t)_3D_TRUSS_THREE_ELEMENTS_8_ANA.py

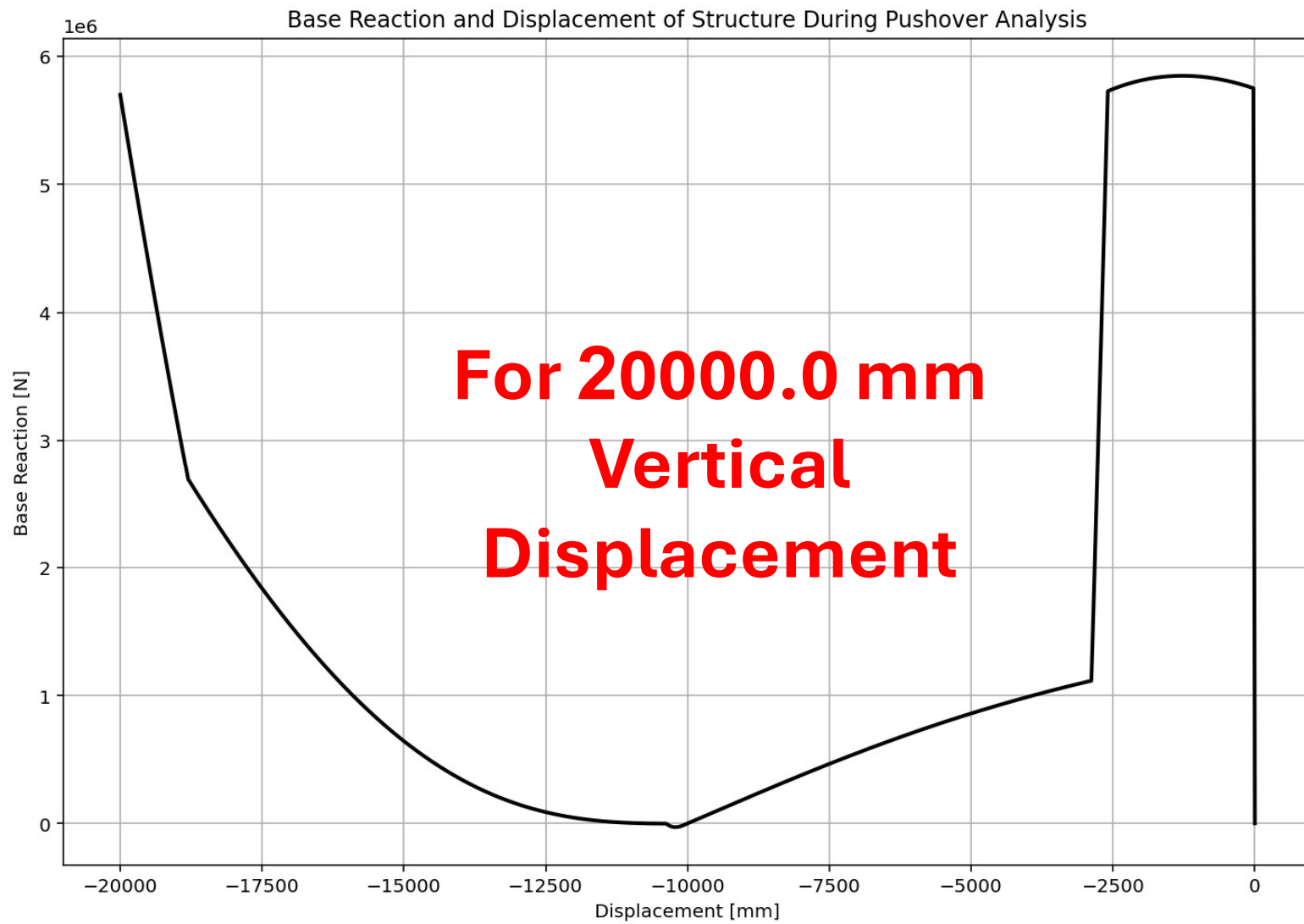
P(t)_3D_TRUSS_THREE_ELEMENTS_8_ANA.py x PLOT_3D_TRUSS.py x

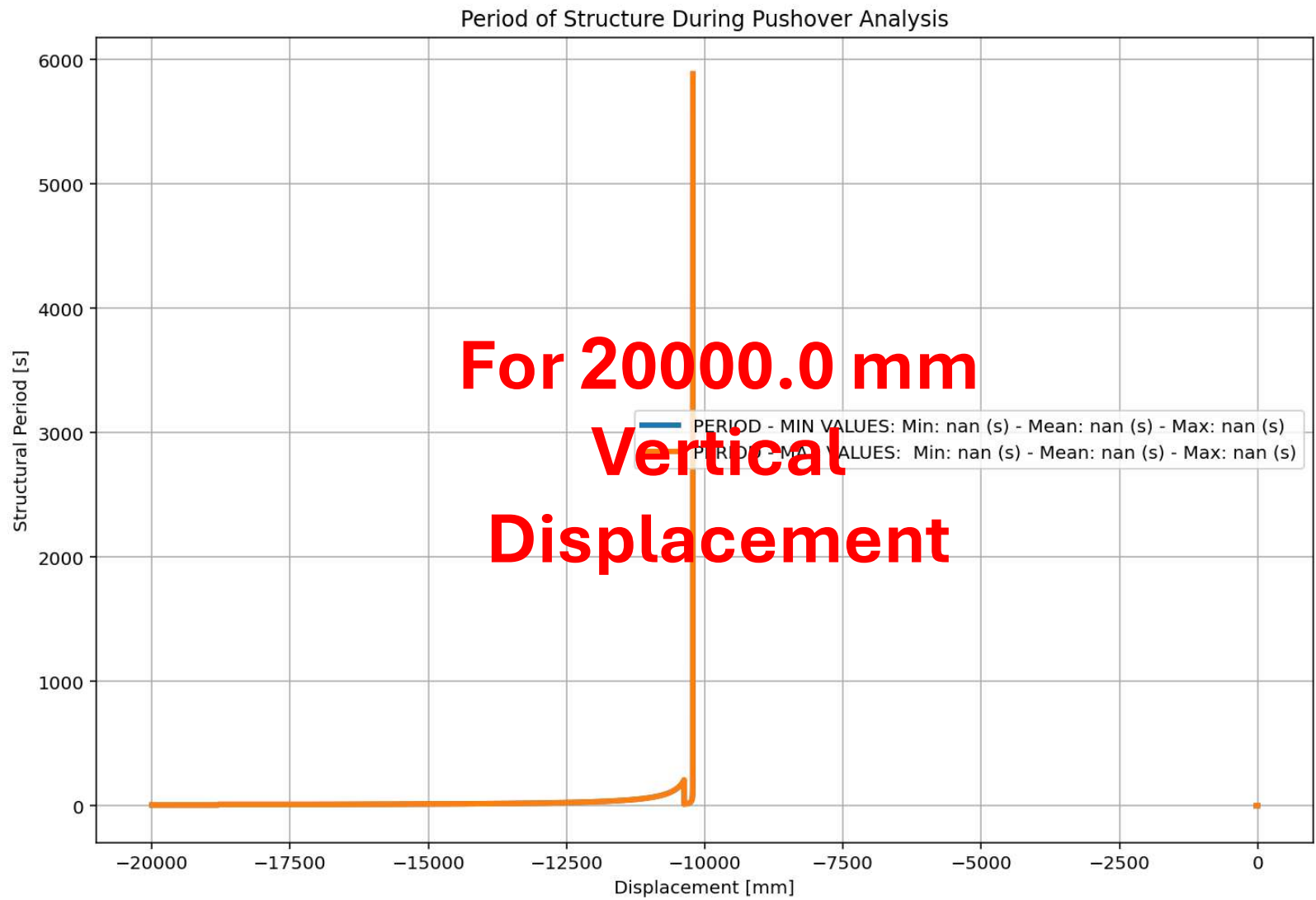
```
299 if ANAL_TYPE == 'PUSHOVER': # STATIC TIME-HISTORY ANALYSIS
300     IDctrlDOF = 3 # 1: Horizontal Displcement - 2: Vertical Displcement
301     DINCR = -0.01 # [mm] Incremental Vertical Displacement
302     DMAX = -100.0 # [mm] Max. Vertical Displacement
303     ops.integrator('DisplacementControl', center_node, IDctrlDOF, DINCR)
304     ops.analysis('Static')
305     Nsteps = int(np.abs(DMAX/ DINCR))
306     STEP = 0.0
307     for step in range(Nsteps):
308         OK = ops.analyze(1)
309         S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
310         ops.reactions()
311         reaction.append(ops.nodeReaction(1, 3)+ops.nodeReaction(3, 3)+ops.nodeReaction(
312             disp_mid.append(ops.nodeDisp(center_node, 3)) # DISPLACEMENT
313             dispX.append(ops.nodeDisp(center_node, 1)) # DISPLACEMENT
314             dispY.append(ops.nodeDisp(center_node, 2)) # DISPLACEMENT
315             dispZ.append(ops.nodeDisp(center_node, 3)) # DISPLACEMENT
316             # Store elements force, strain and stress
317             for ele_id in ele_axialforce.keys():
318                 ele_axialforce[ele_id].append(ops.eleResponse(ele_id, 'axialForce')) #
319                 ele_stress[ele_id].append(ops.eleResponse(ele_id, 'material', 'stress')) #
320                 ele_strain[ele_id].append(ops.eleResponse(ele_id, 'material', 'strain')) #
321                 if ELE_TYPE == 'INELASTIC':
322                     ele_di[ele_id].append(100*(abs(ele_strain[ele_id]) - ey) - (esu - ey))
323                 if ELE_TYPE == 'ELASTIC':
324                     ele_di[ele_id].append(0.0)
325             # Store displacements
326             for node_id in node_displacementsX.keys():
327                 node_displacementsX[node_id].append(ops.nodeDisp(node_id, 1))
328                 node_displacementsY[node_id].append(ops.nodeDisp(node_id, 2))
329                 node_displacementsZ[node_id].append(ops.nodeDisp(node_id, 3))
330             # IN EACH STEP, STRUCTURE PERIOD GOING TO BE CALCULATED
331             #PERIODmin, PERIODmax = S06.RAYLEIGH_DAMPING(1, 0.5*DR, DR, 0, 1)
332             PERIODmin, PERIODmax = S05.EIGENVALUE_ANALYSIS(1, PLOT=True)
```

Base Reaction and Displacement of Structure During Pushover Analysis

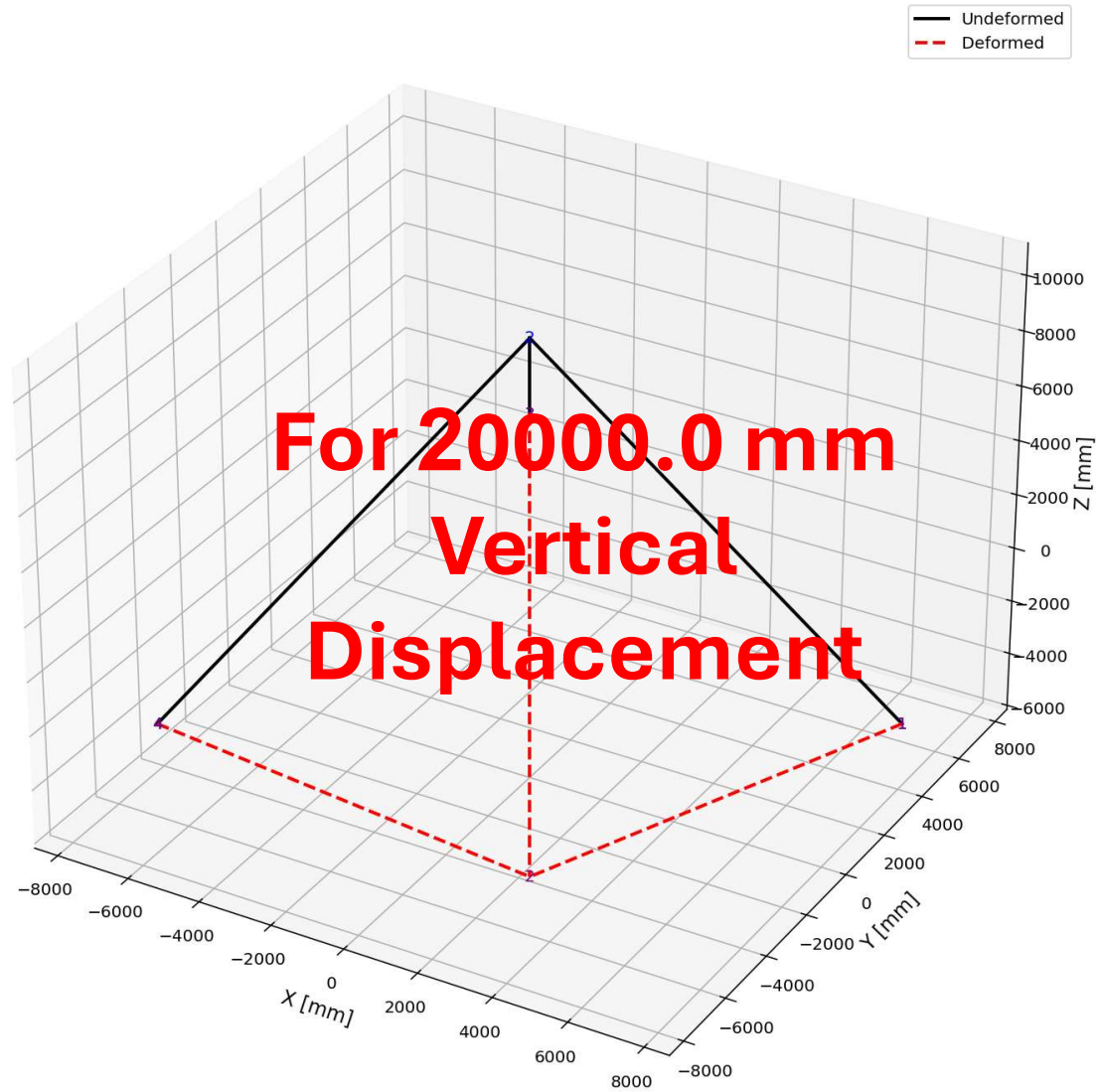
Python Console Files Help Variable Explorer Debugger Plots History

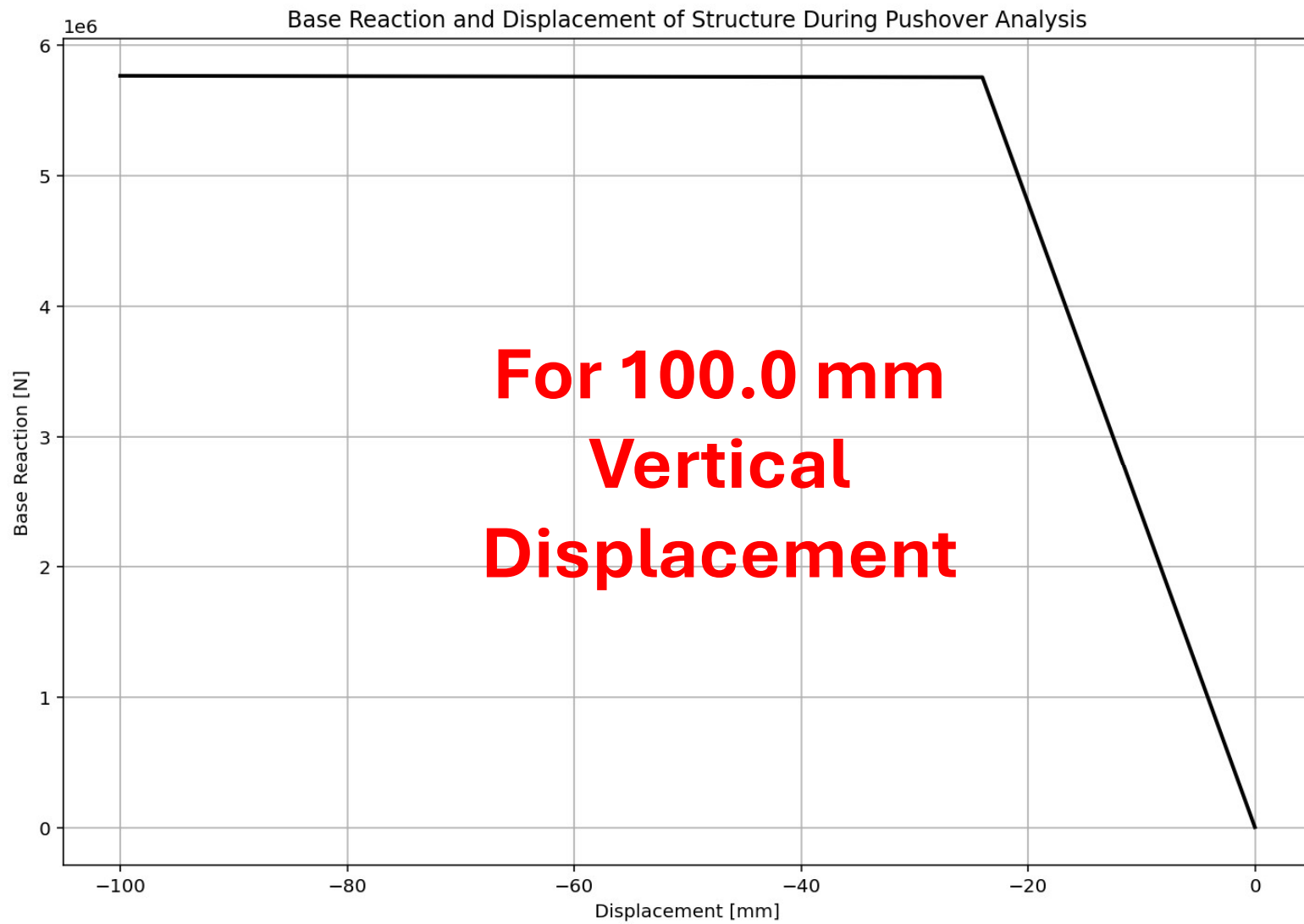
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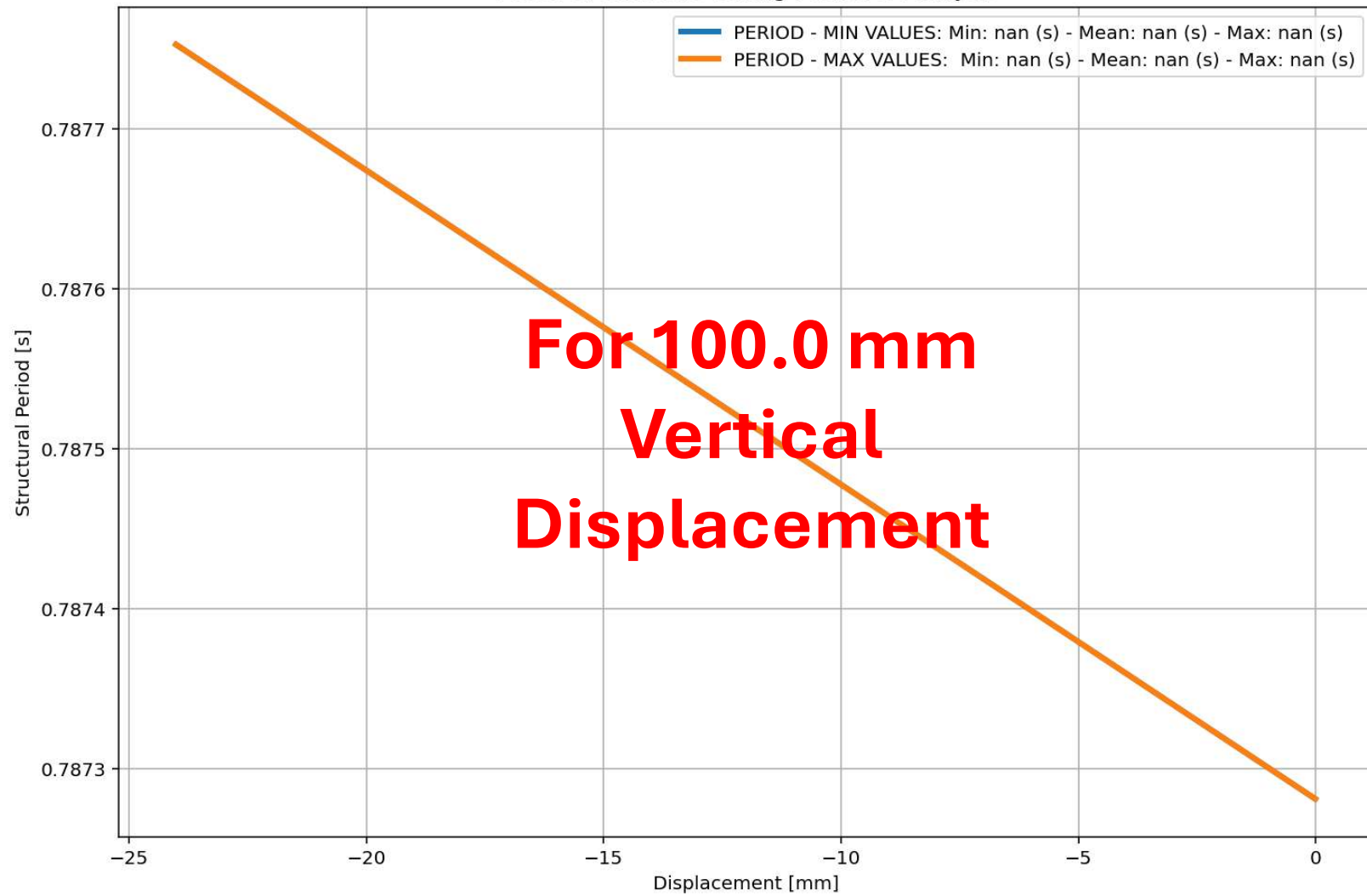


3D Truss: Undeformed vs Deformed (Scale = 1.00)

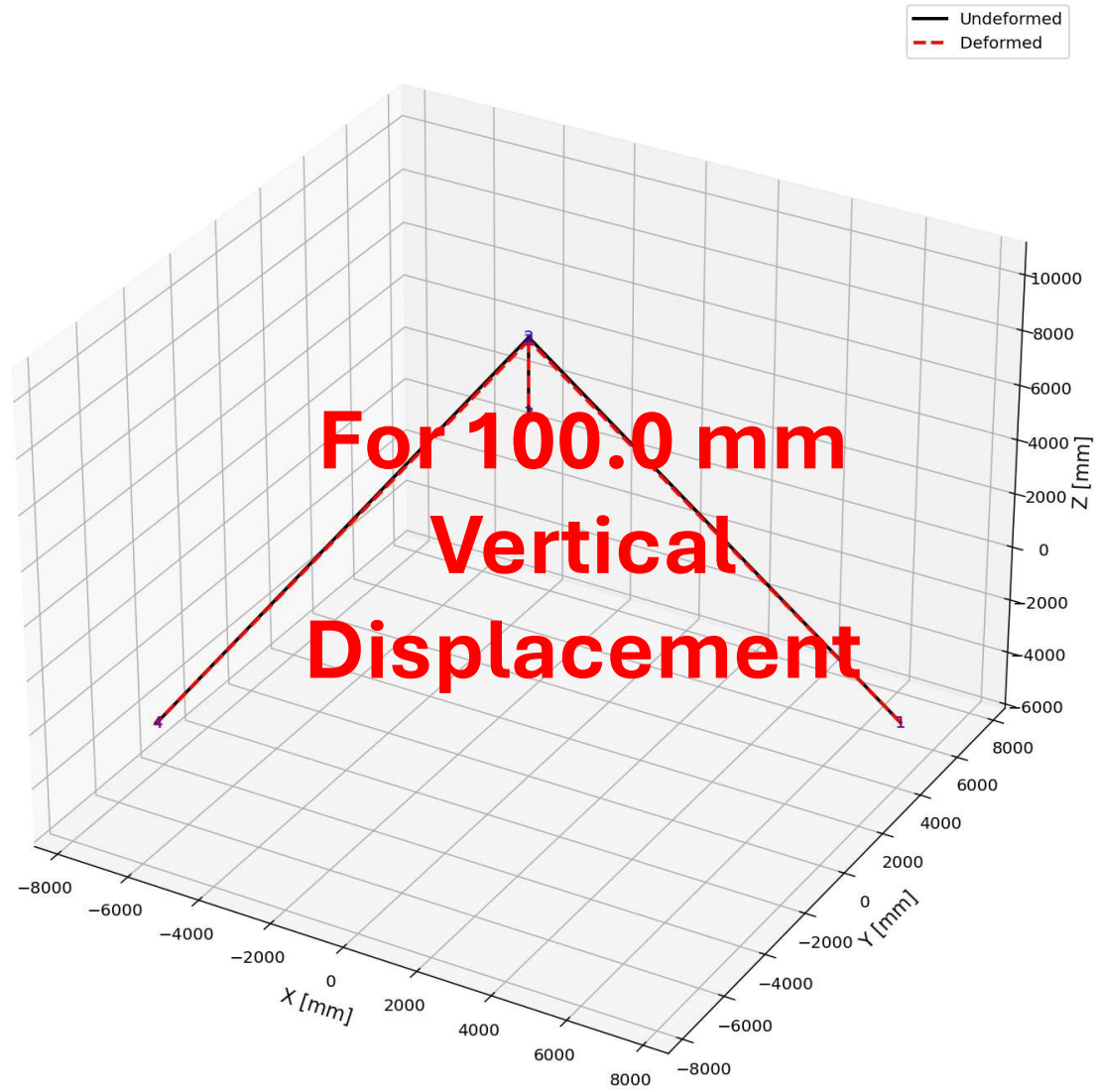




Period of Structure During Pushover Analysis



3D Truss: Undeformed vs Deformed (Scale = 1.00)



CYCLIC DISPLACEMENT ANALYSIS

Spyder (Python 3.12)

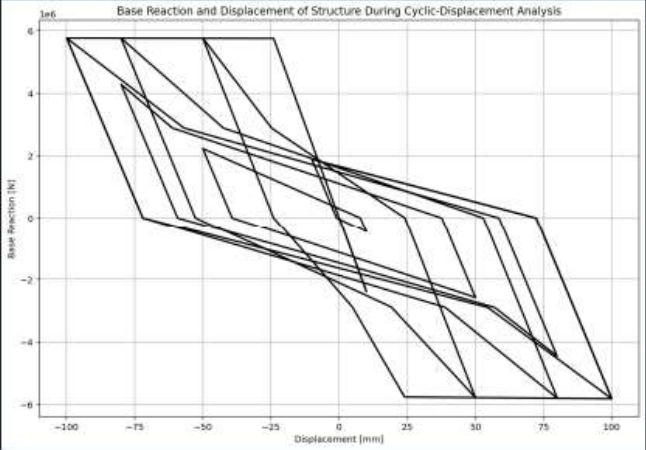
File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\HEIGHT_ANALY...MENTS_8_ANA\P(t)_3D_TRUSS_THREE_ELEMENTS_8_ANA.py

P(t)_3D_TRUSS_THREE_ELEMENTS_8_ANA.py X PLOT_3D_TRUSS.py X

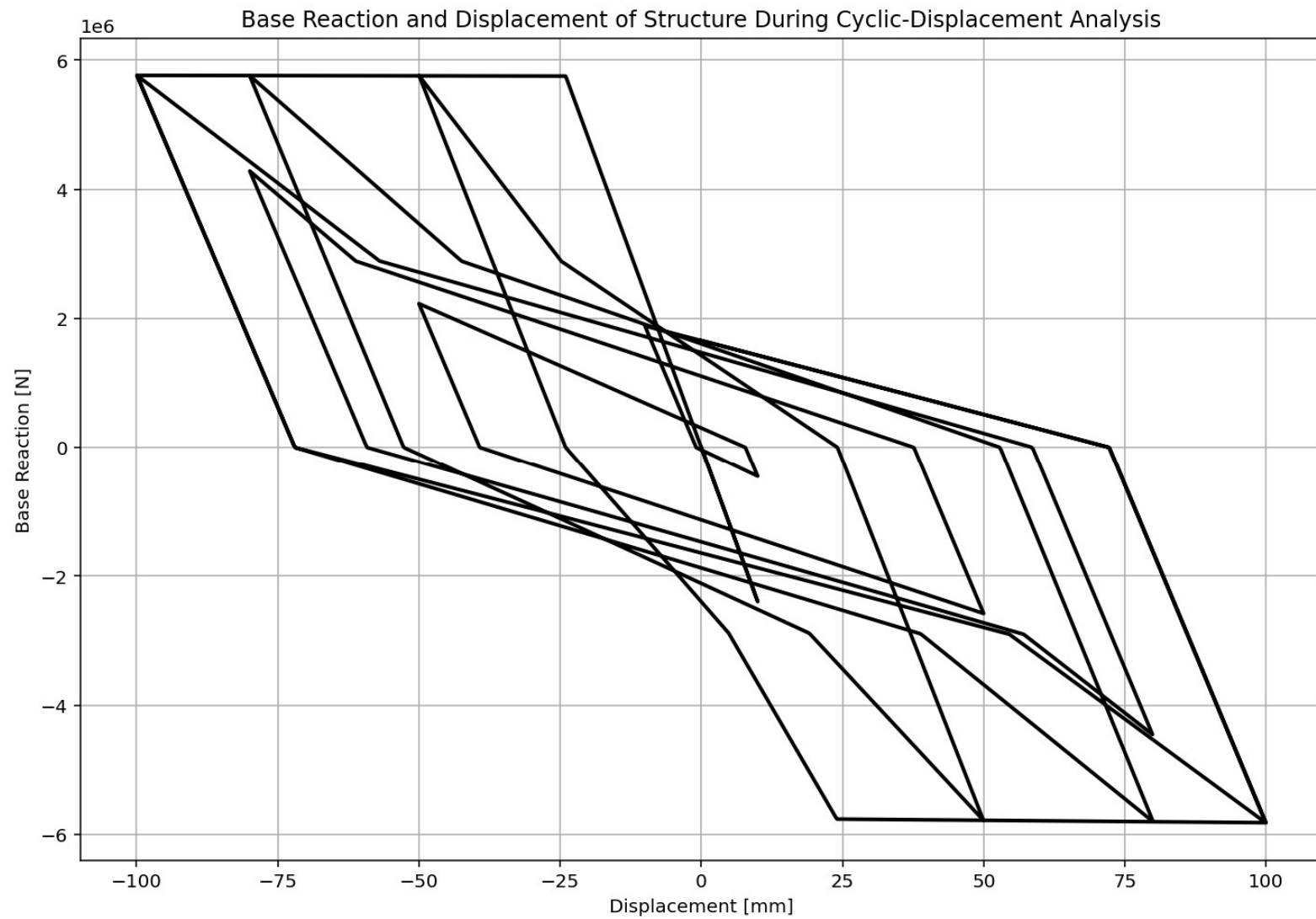
```
348
349 if ANAL_TYPE == 'CYCLIC_DISPLACEMENT': # STATIC TIME-HISTORY ANALYSIS
350     IDctrlDOF = 3 # 1: Horizontal Displacement - 2: Vertical Displacement
351     DMAX = -100.0 # [mm] Max. Vertical Displacement
352     n_points = 10000
353     # 4. CYCLIC DISPLACEMENT PROTOCOL
354     # Key strain points (same logic as your protocol)
355     key_disp = np.array([
356         0.0,
357         0.1*DMAX, -0.1*DMAX,
358         0.5*DMAX, -0.5*DMAX,
359         0.8*DMAX, -0.8*DMAX,
360         DMAX, -DMAX,
361         0.1*DMAX, -0.1*DMAX,
362         0.5*DMAX, -0.5*DMAX,
363         0.8*DMAX, -0.8*DMAX,
364         DMAX, -DMAX,
365         0.0
366     ])
367
368     # Generate 1000-point displacement protocol
369     disp_protocol = np.interp(
370         np.linspace(0, len(key_disp) - 1, n_points),
371         np.arange(len(key_disp)),
372         key_disp
373     )
374
375     ops.analysis('Static')
376     STEP = 0.0
377     for target_disp in disp_protocol:
378         current_disp = ops.nodeDisp(center_node, 3) # DISPALCEMENT APPLIED IN MIDDLE N
379         dU = target_disp - current_disp
380         ops.integrator('DisplacementControl', center_node, IDctrlDOF, dU)
381         OK = ops.analyze(1)
```

Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis

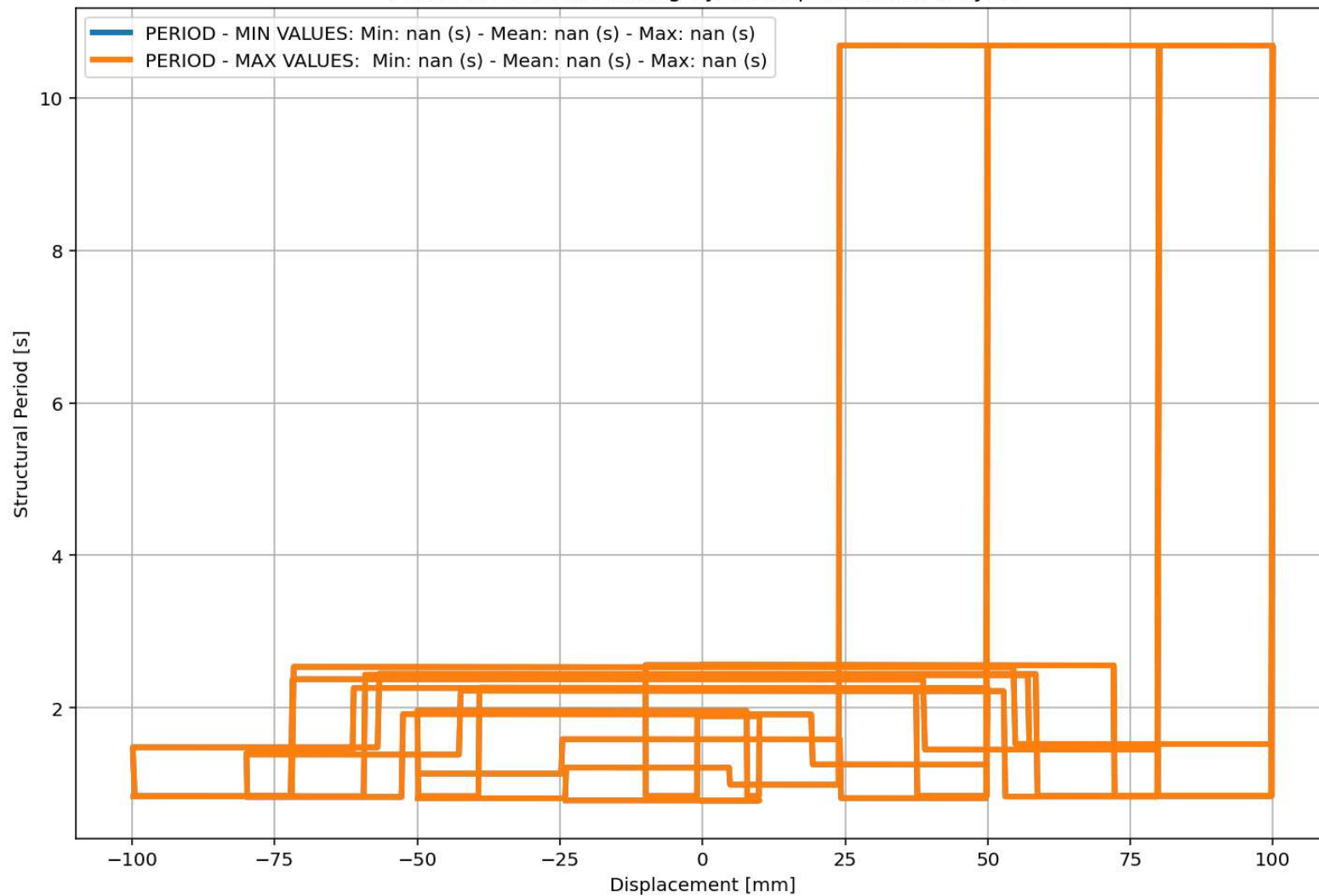


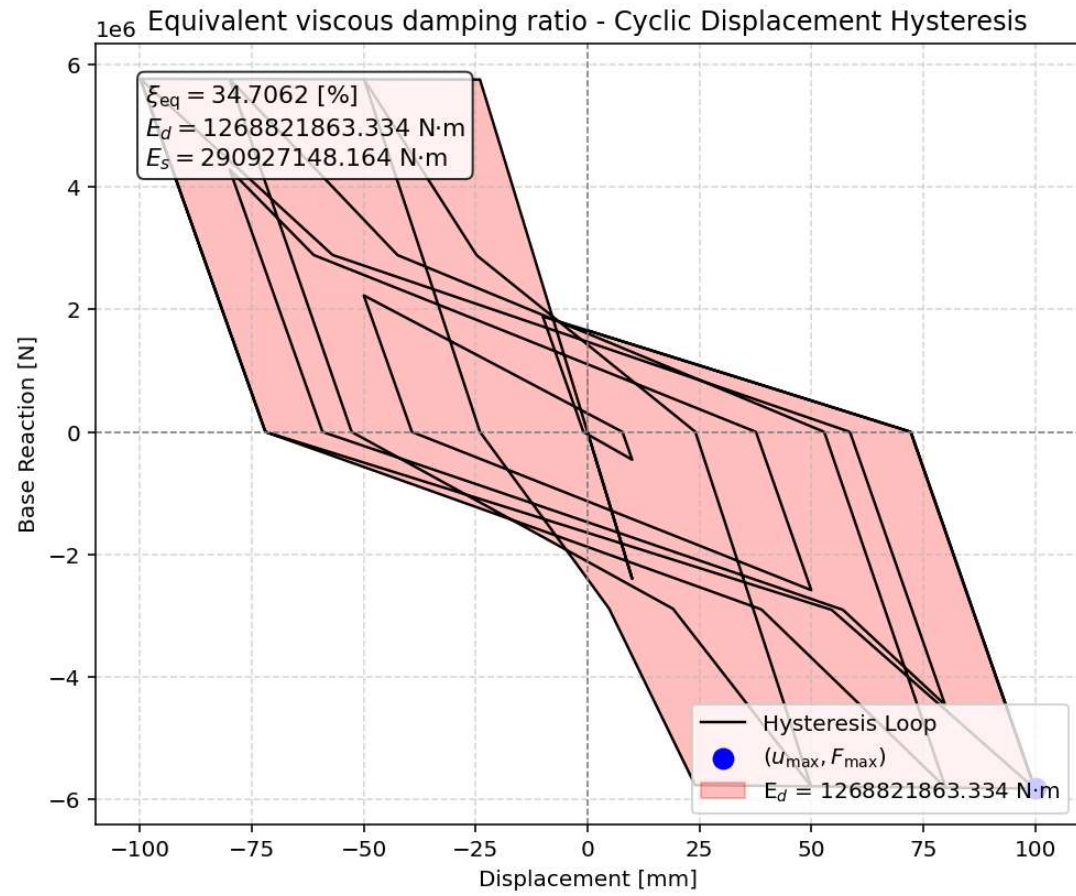
[Python Console] Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 33%

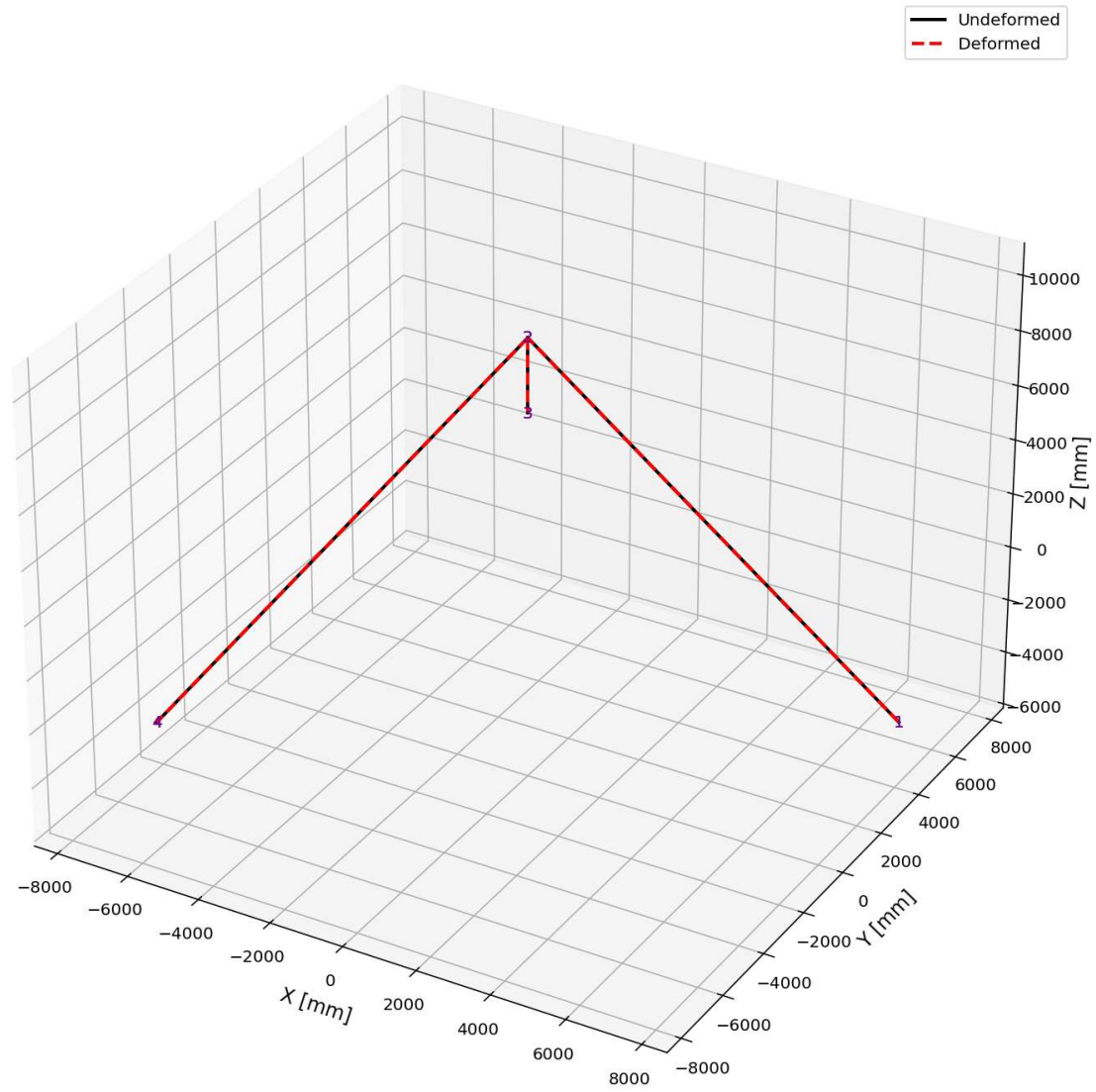


Period of Structure During Cyclic Displacement Analysis





3D Truss: Undeformed vs Deformed (Scale = 1.00)



STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

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C:\Users\De\l\Desktop\OPENSEES_FILES\+EIGHT_ANALY...\26_TRUSS_TWO_ELEMENTS\P(t)_TRUSS_TWO_ELEMENTS.py

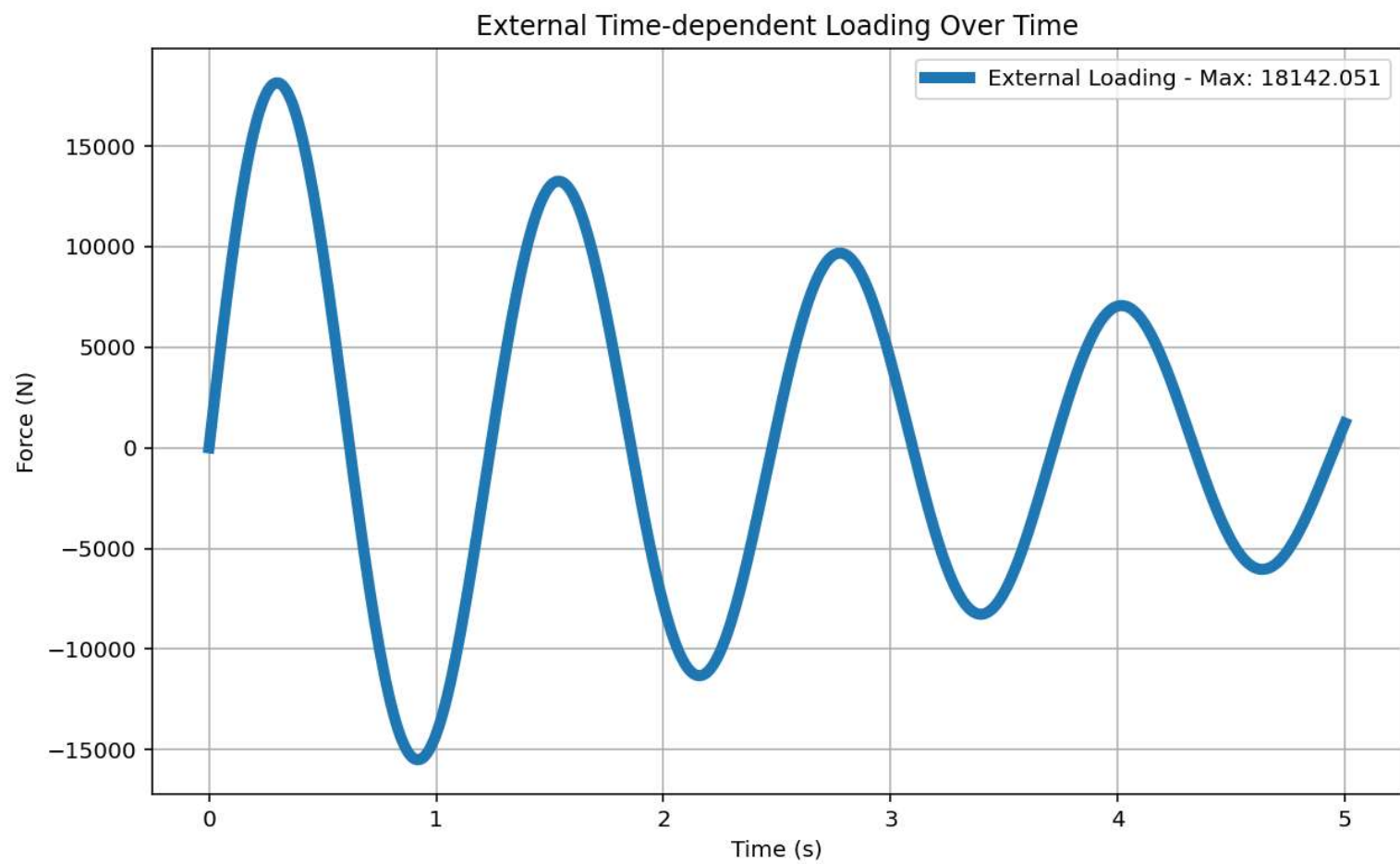
P(t)_TRUSS_TWO_ELEMENTS.py

```
398 if ANAL_TYPE == 'STATIC EXTERNAL TIME-DEPENDENT LOADING': # STATIC TIME-HISTORY ANALYSIS
399     %% DEFINE EXTERNAL TIME-DEPENDENT LOADING PROPERTIES
400     # IN HERE ANALYSIS TIME AND DURATION ARE LOAD STEPS
401     duration = 20.0 # [s] Analysis duration
402     dt = 0.01 # [s] Time step
403     DT = dt # [s] Time step
404     DT_time = 5.0 # [s] Total external Load Analysis Durations [*****]
405     force_amplitude = 19600.0 # [N] Amplitude Force
406     omega_DT = 5.0715 # [rad/s] Natural angular frequency
407     time_steps = int(duration/dt)
408
409     # Check function
410     def CHECK_FUN(DT_time ,duration):
411         if DT_time > duration:
412             print('\n\nAnalysis Duration Must be greater than External Load Duration\n')
413             exit()
414         return -1
415
416     CHECK_FUN(DT_time ,duration)
417     def EXTERNAL_TIME_DEPENDENT(force_amplitude, omega_DT, DT, DT_time): # P(t) = P0 sin
418         import numpy as np
419         import matplotlib.pyplot as plt
420         # External Load Durations
421         num_steps = int(DT_time / DT)
422         load_time = np.linspace(0, DT_time, num_steps)
423         target_frequency = 1.0 * omega_DT # Target excitation frequency
424         DT_load = force_amplitude * np.sin(target_frequency * load_time)
425         # Plot External Time-dependent Loading
426         plt.figure(figsize=(10, 6))
427         plt.plot(load_time, DT_load, label=f'External Loading - Max: {np.max(DT_load)}')
428         plt.title('External Time-dependent Loading Over Time')
429         plt.xlabel('Time (s)')
430         plt.ylabel('Force (N)')
431         plt.grid(True)
```

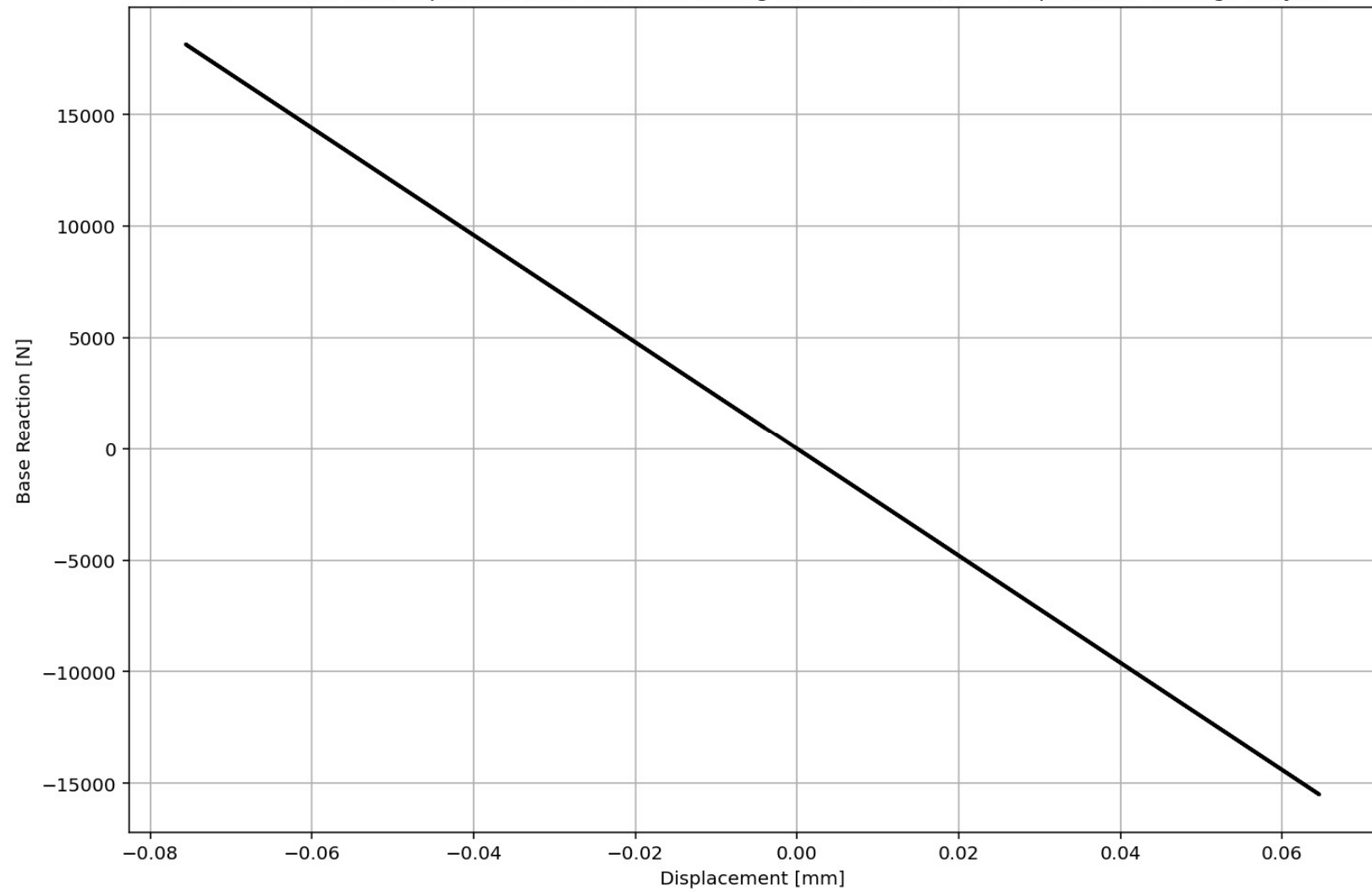
Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis

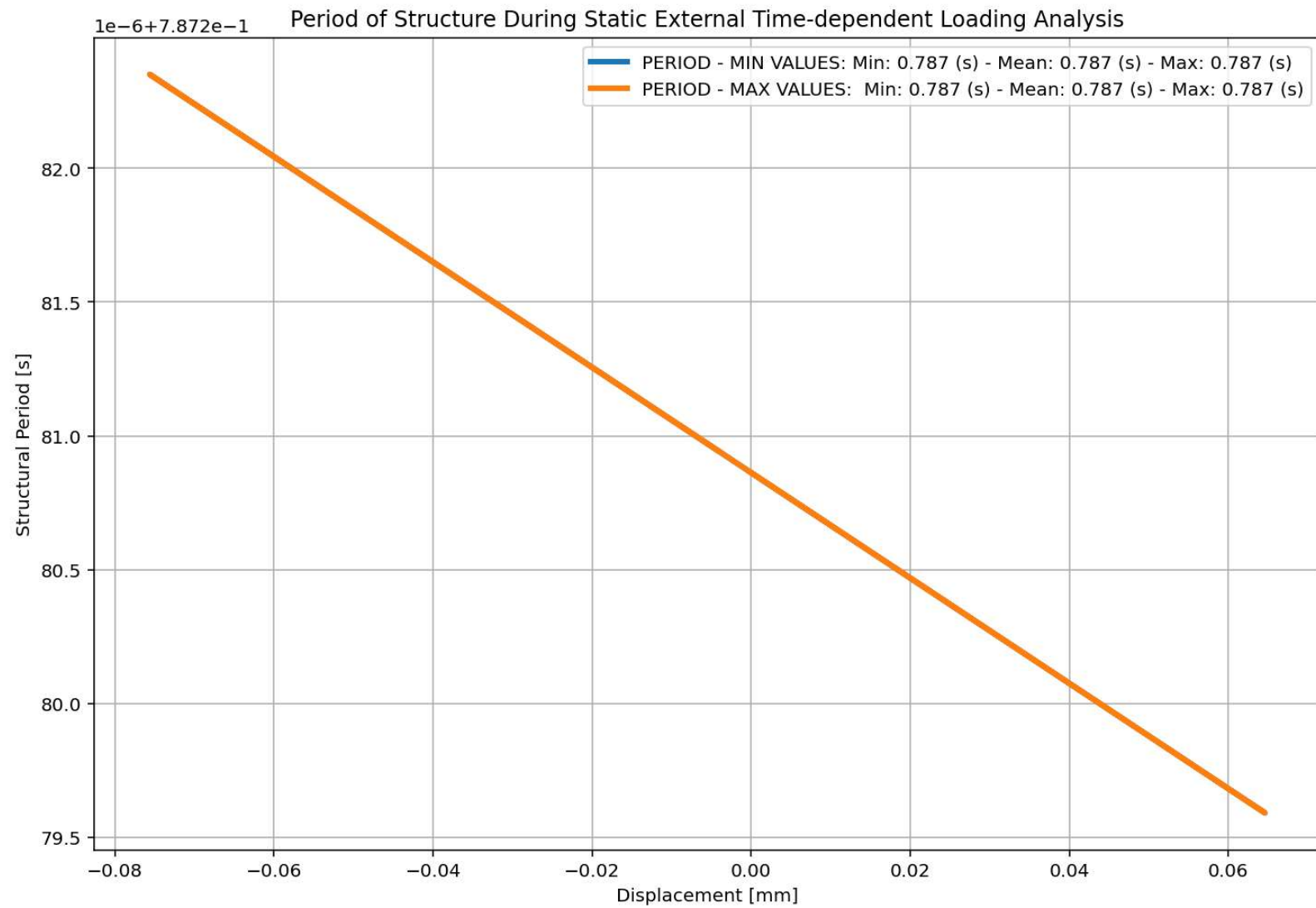
Python Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 7, Col 44 UTF-8 CRLF RW Mem 50%

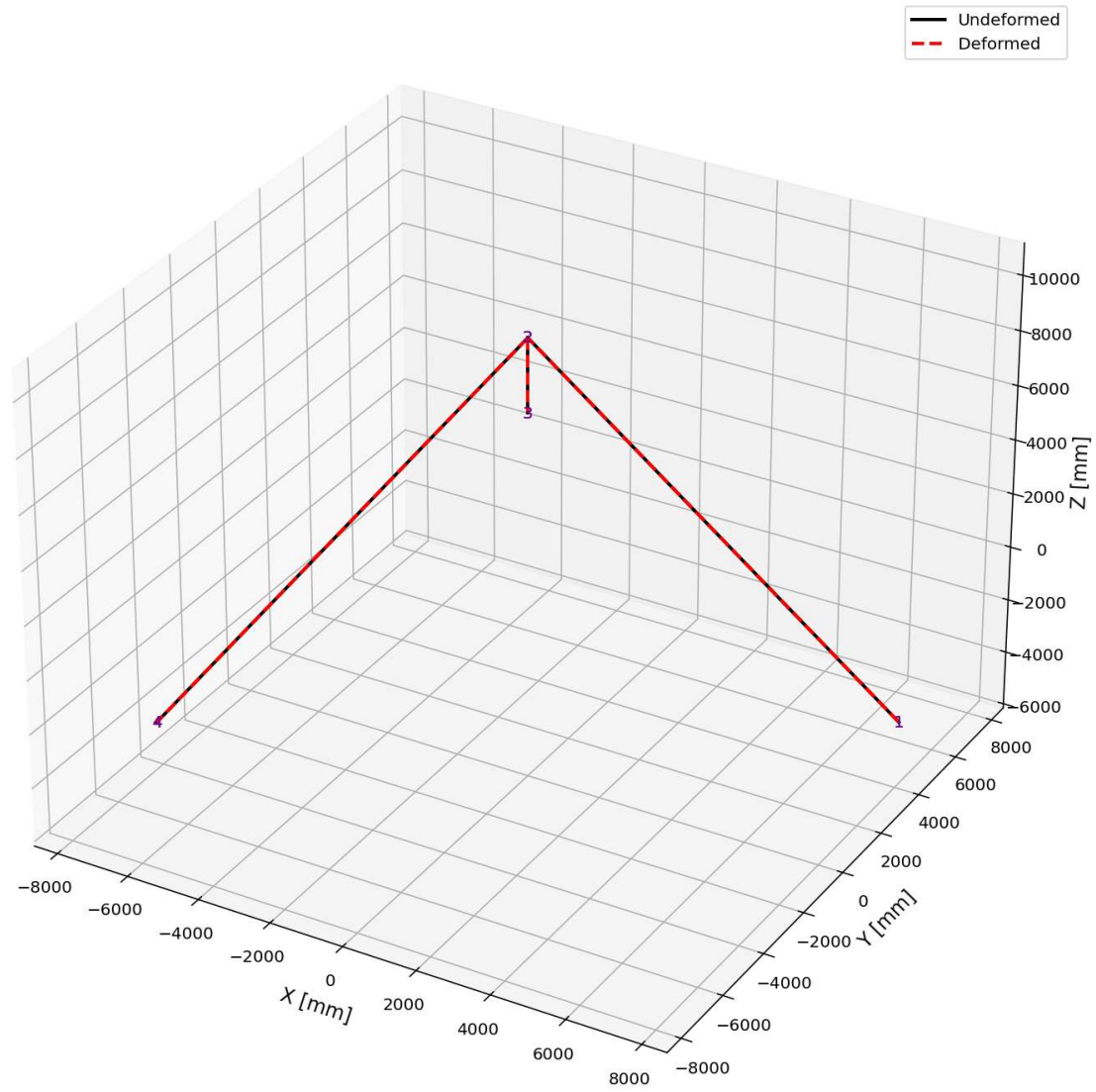


Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis





3D Truss: Undeformed vs Deformed (Scale = 1.00)



DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

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P(t)_3D_TRUSS_THREE_ELEMENTS_8_ANA.py x PLOT_3D_TRUSS.py x

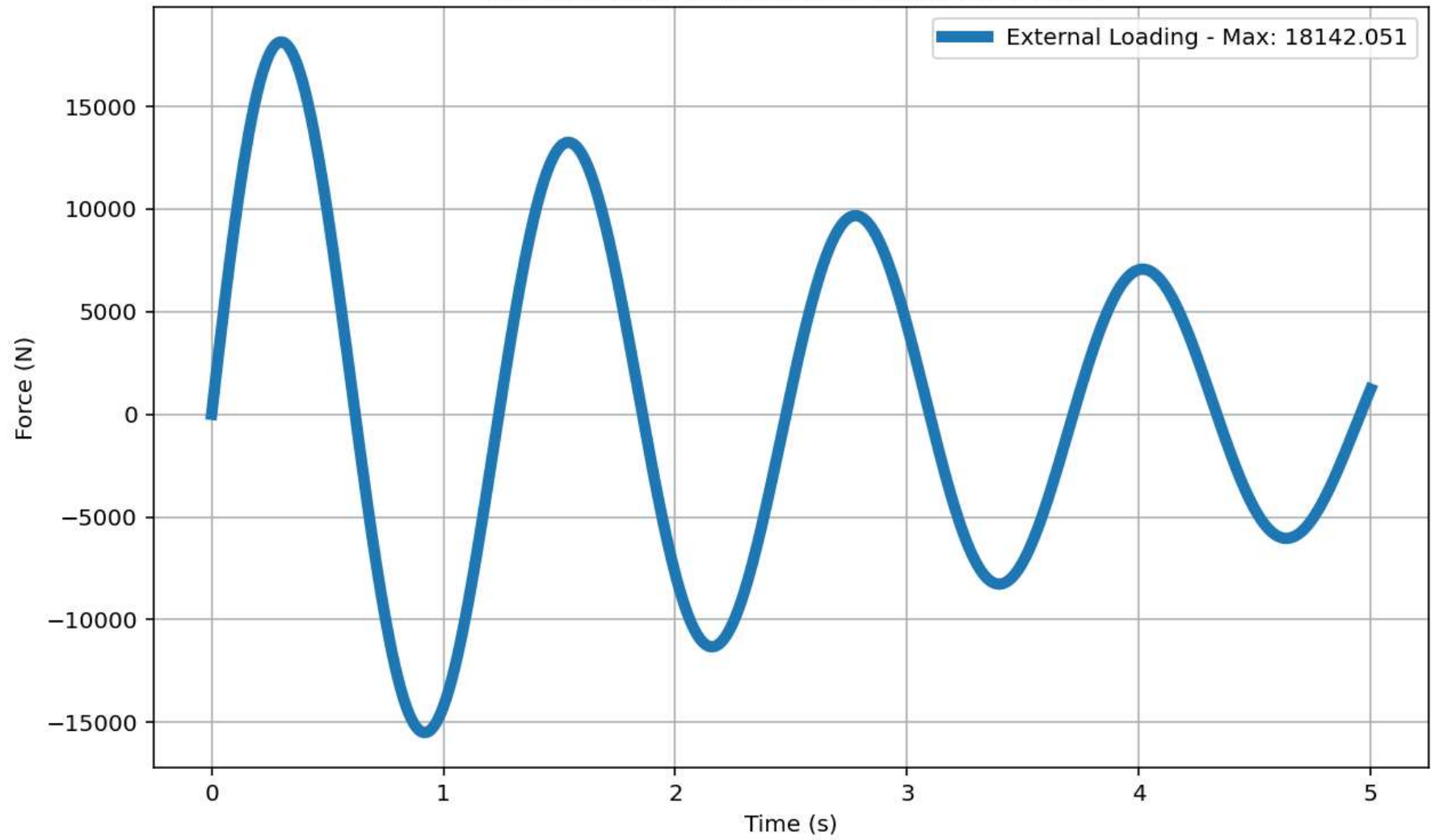
```
549
550 if ANAL_TYPE == 'DYNAMIC EXTERNAL TIME-DEPENDENT LOADING': # DYNAMIC TIME-HISTORY ANAL
551     %% DEFINE EXTERNAL TIME-DEPENDENT LOADING PROPERTIES
552     duration = 20.0 # [s] Analysis duration
553     dt = 0.01 # [s] Time step
554     DT = dt # [s] Time step
555     DT_time = 5.0 # [s] Total external Load Analysis Durations [*****]
556     force_amplitude = 19600.0 # [N] Amplitude Force
557     omega_DT = 5.0715 # [rad/s] Natural angular frequency
558     DR = 0.03 # Damping Ratio
559
560 # Check function
561 def CHECK_FUN(DT_time ,duration):
562     if DT_time > duration:
563         print('\n\nAnalysis Duration Must be greater than External Load Duration\n')
564         exit()
565     return -1
566
567 CHECK_FUN(DT_time ,duration)
568 def EXTERNAL_TIME_DEPENDENT(force_amplitude, omega_DT, DT, DT_time): # P(t) = P0 sin
569     import numpy as np
570     import matplotlib.pyplot as plt
571     # External Load Durations
572     num_steps = int(DT_time / DT)
573     load_time = np.linspace(0, DT_time, num_steps)
574     target_frequency = 1.0 * omega_DT # Target excitation frequency
575     DT_load = force_amplitude * np.sin(target_frequency * load_time)
576     # Plot External Time-dependent Loading
577     plt.figure(figsize=(10, 6))
578     plt.plot(load_time, DT_load, label=f'External Loading - Max: {np.max(DT_load)}')
579     plt.title('External Time-dependent Loading Over Time')
580     plt.xlabel('Time (s)')
581     plt.ylabel('Force (N)')
582     plt.grid(True)
```

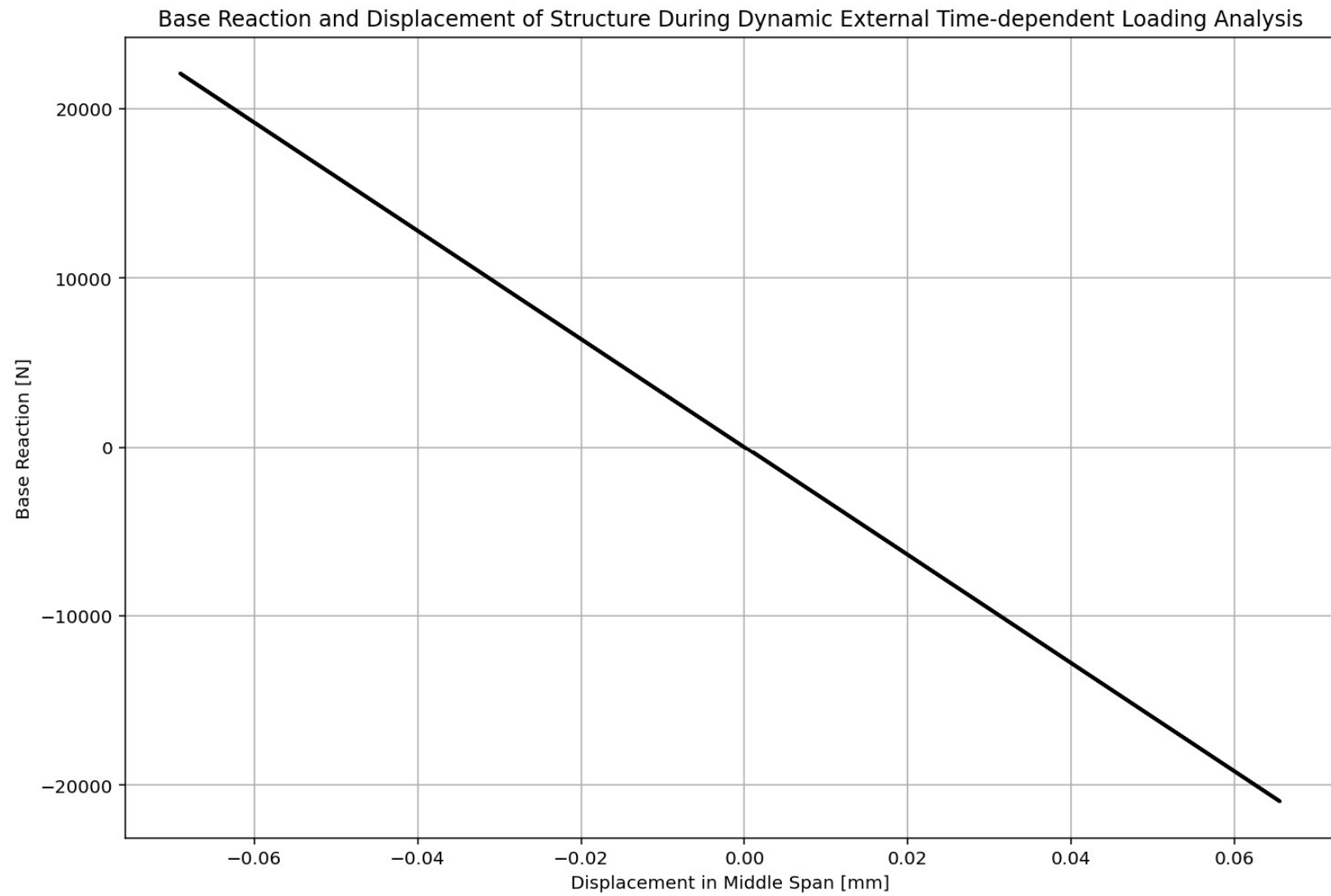
Base Reaction and Displacement of Structure During Free-vibration Analysis

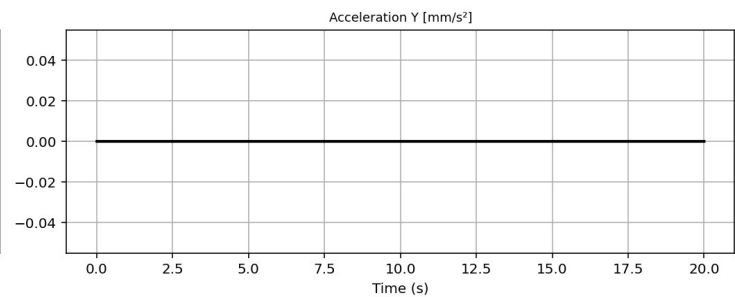
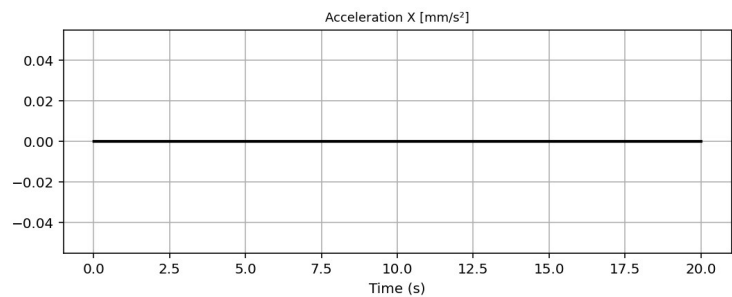
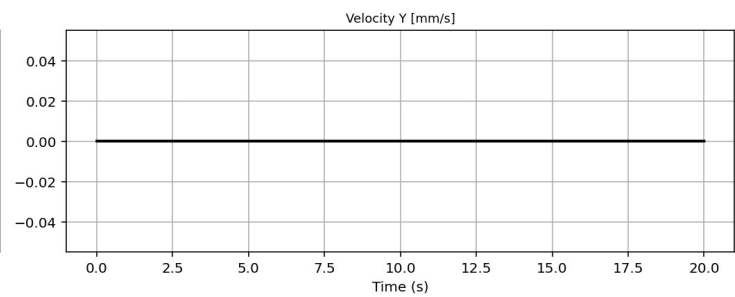
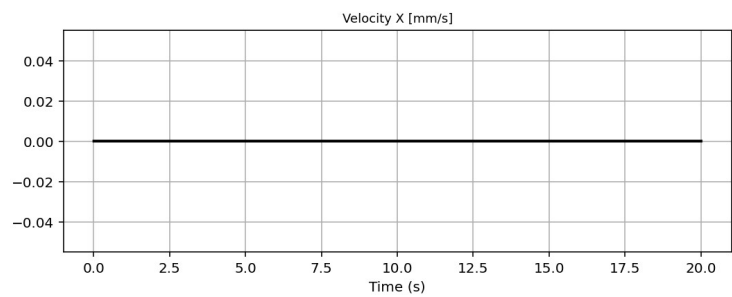
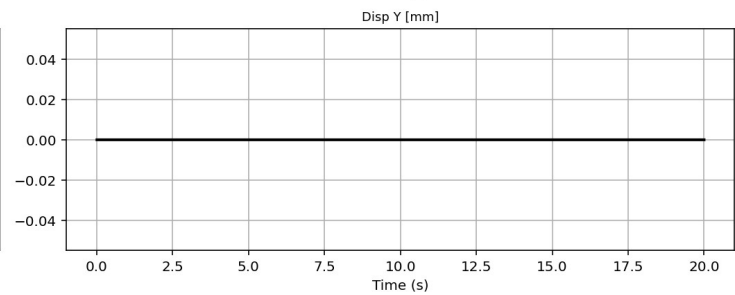
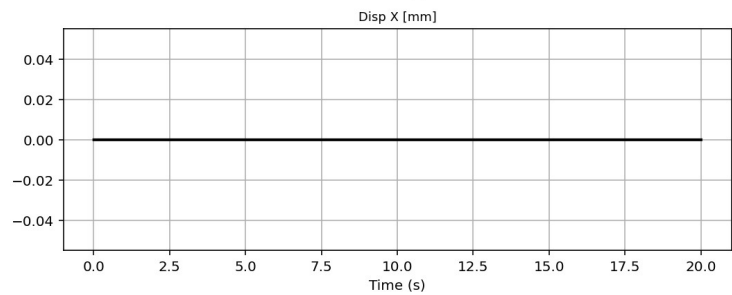
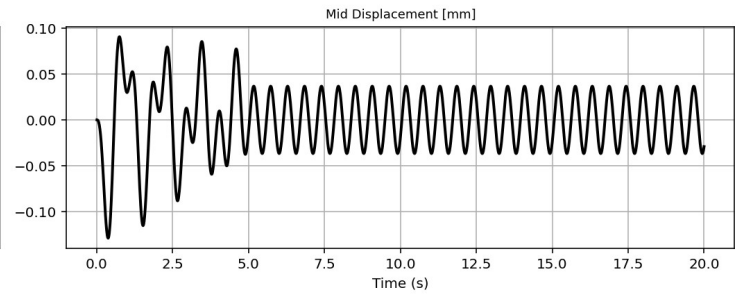
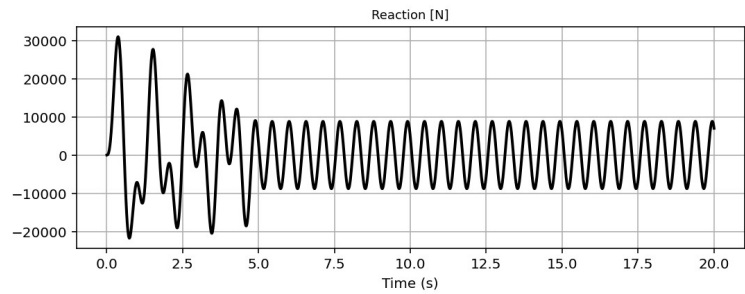
IPython Console Files Help Variable Explorer Debugger Plots History

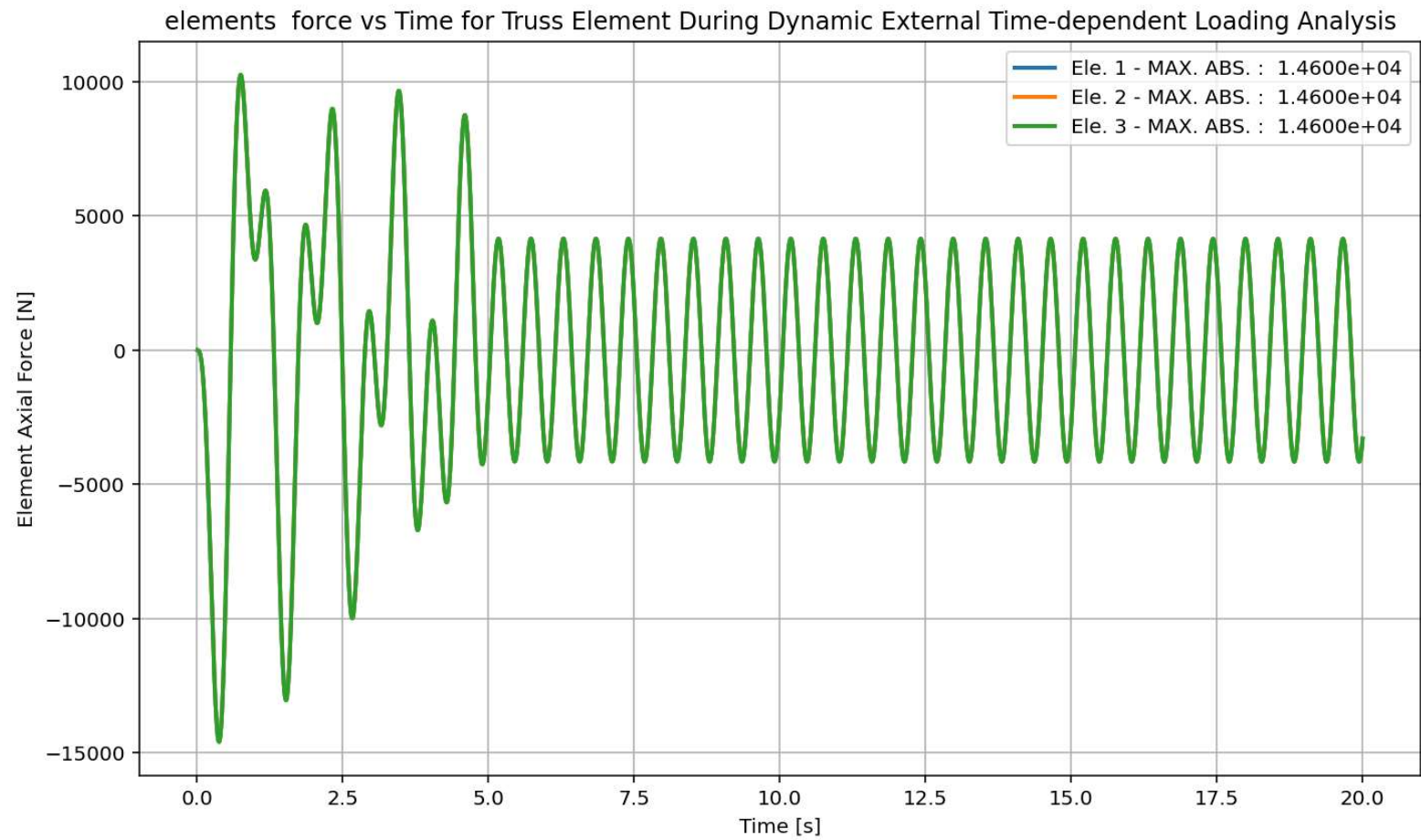
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 302, Col 27 UTF-8 CRLF RW Mem 55%

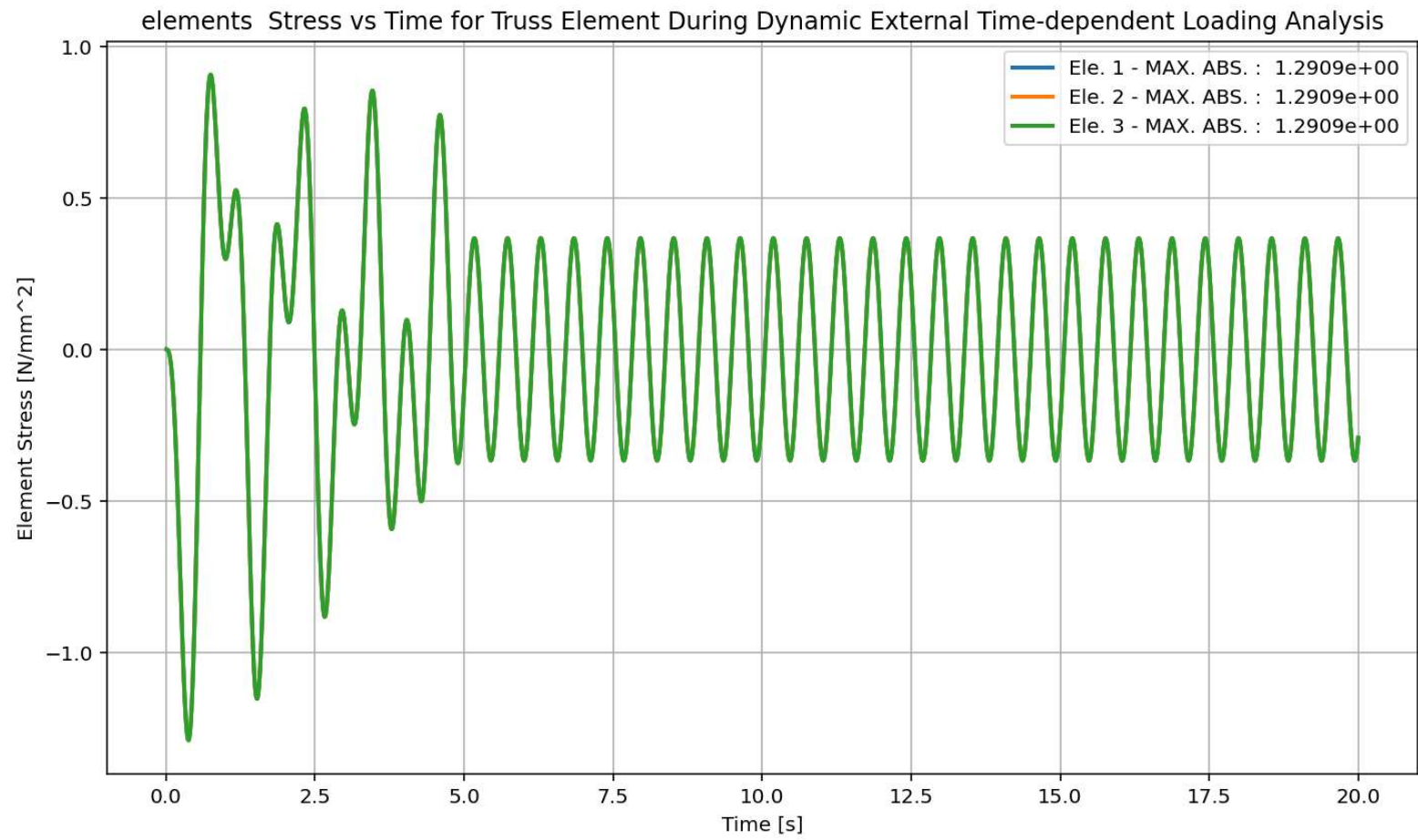
External Time-dependent Loading Over Time



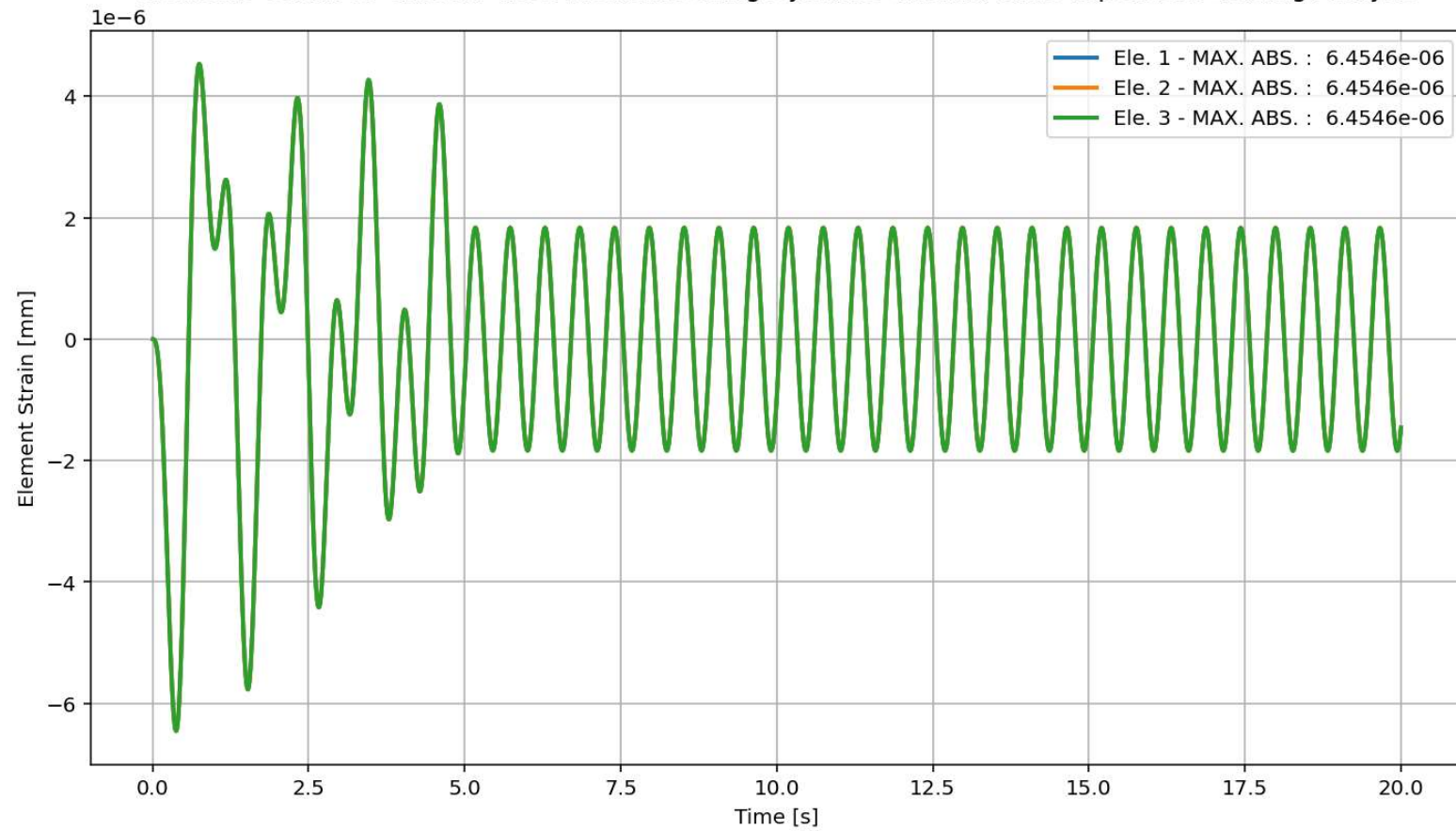




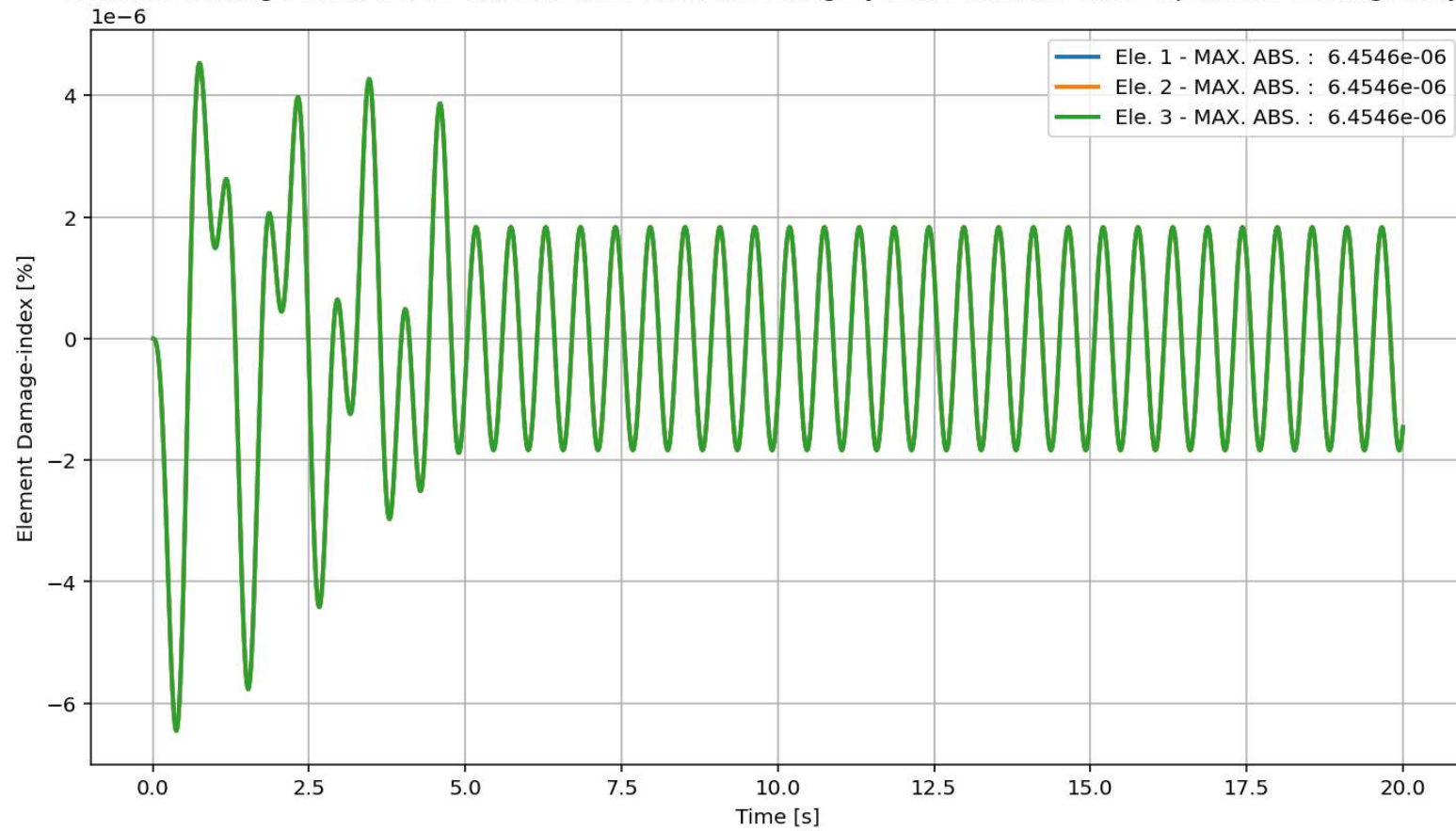




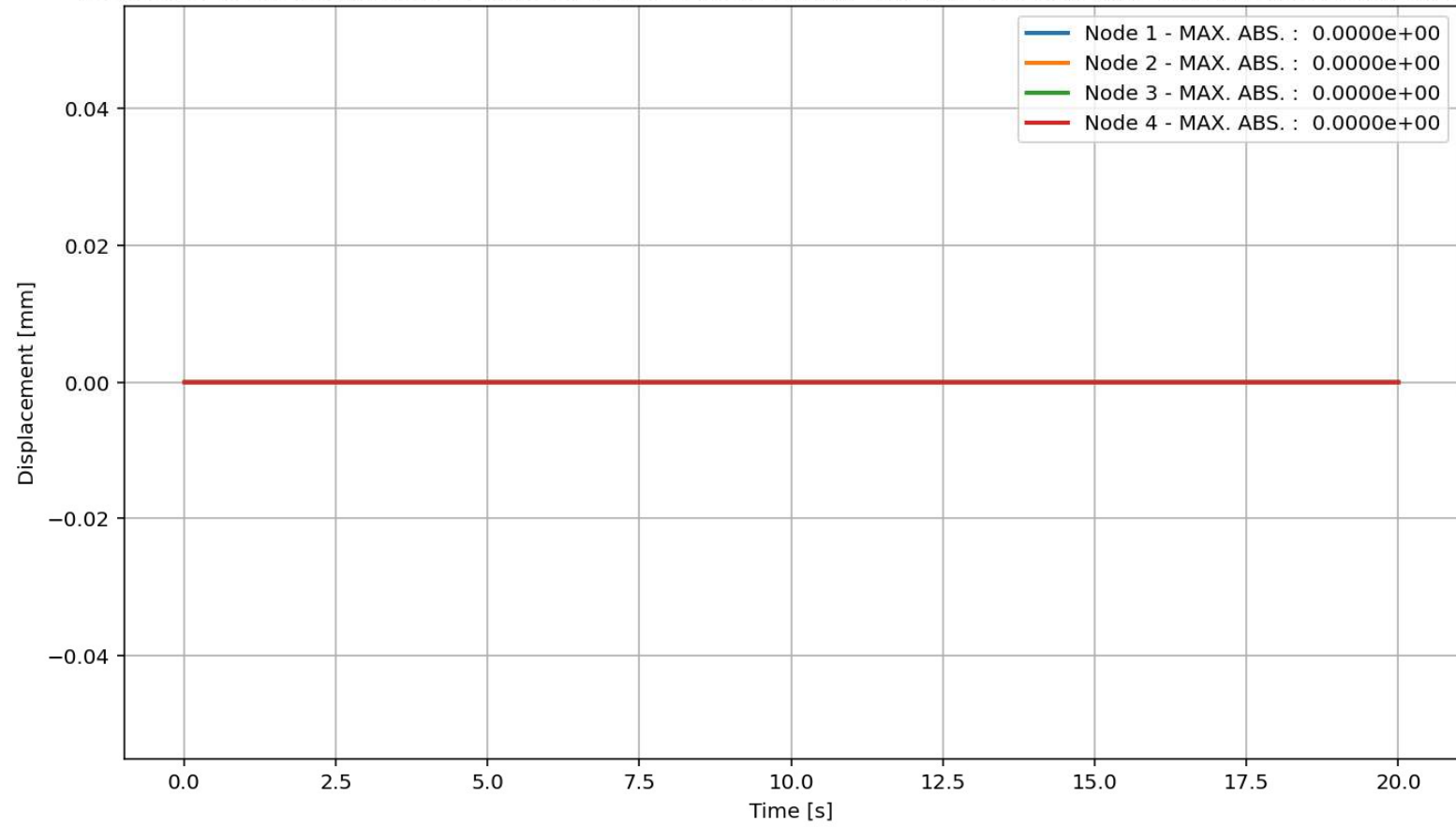
elements Strain vs Time for Truss Element During Dynamic External Time-dependent Loading Analysis



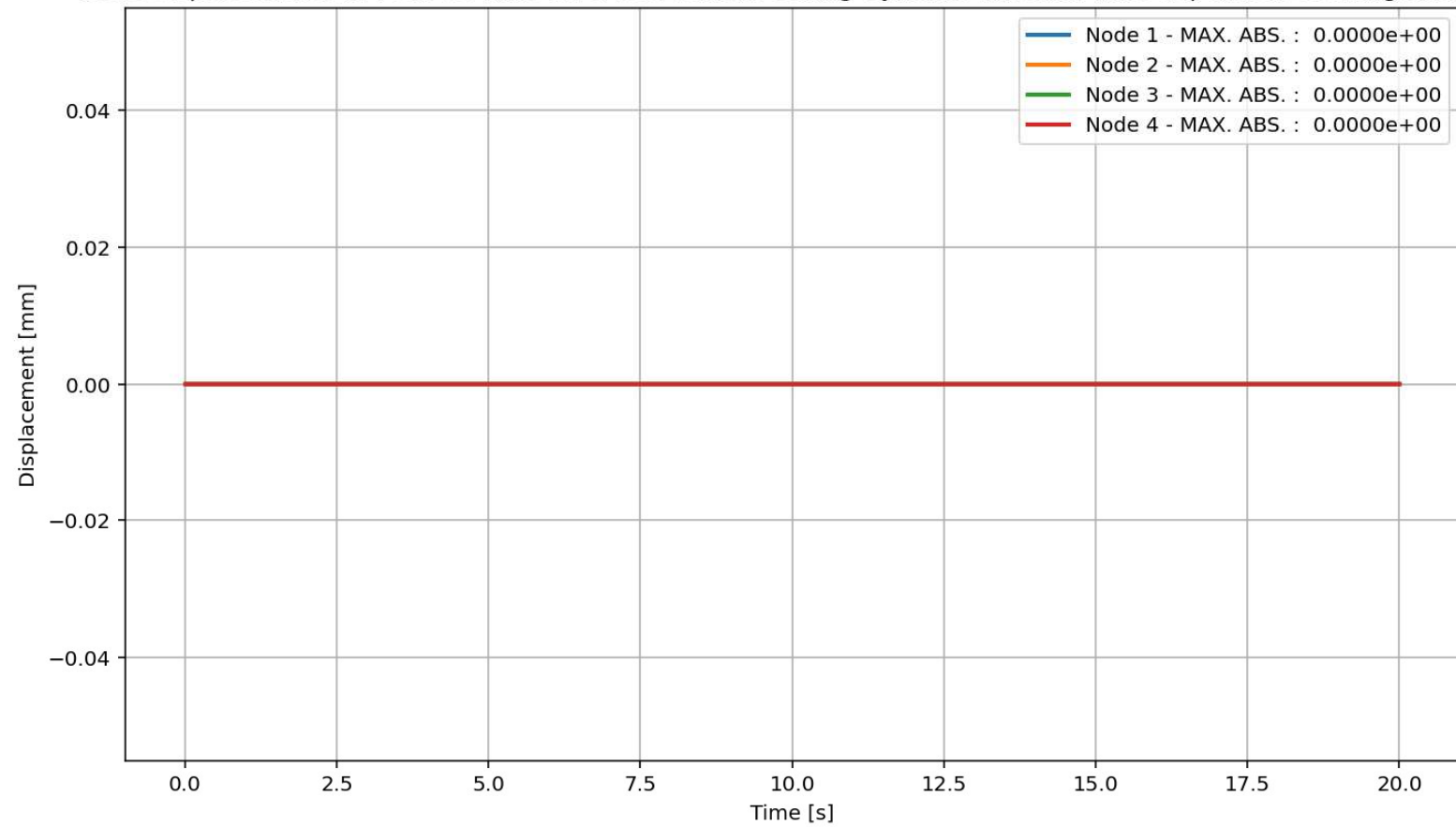
elements Damage-index [%] vs Time for Truss Element During Dynamic External Time-dependent Loading Analysis

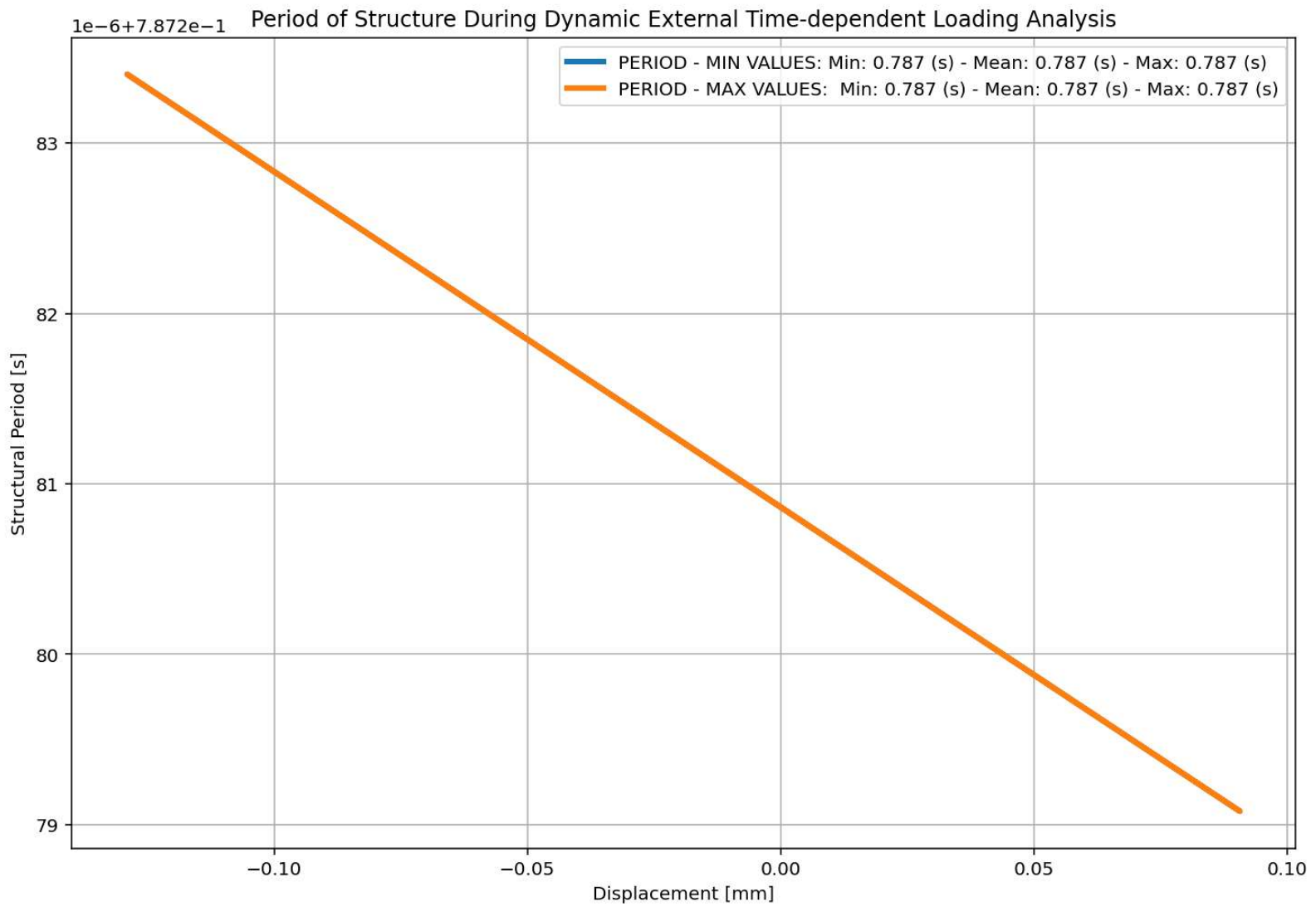


Node Displacements in X Dir. vs Time for Truss Element During Dynamic External Time-dependent Loading Analysis

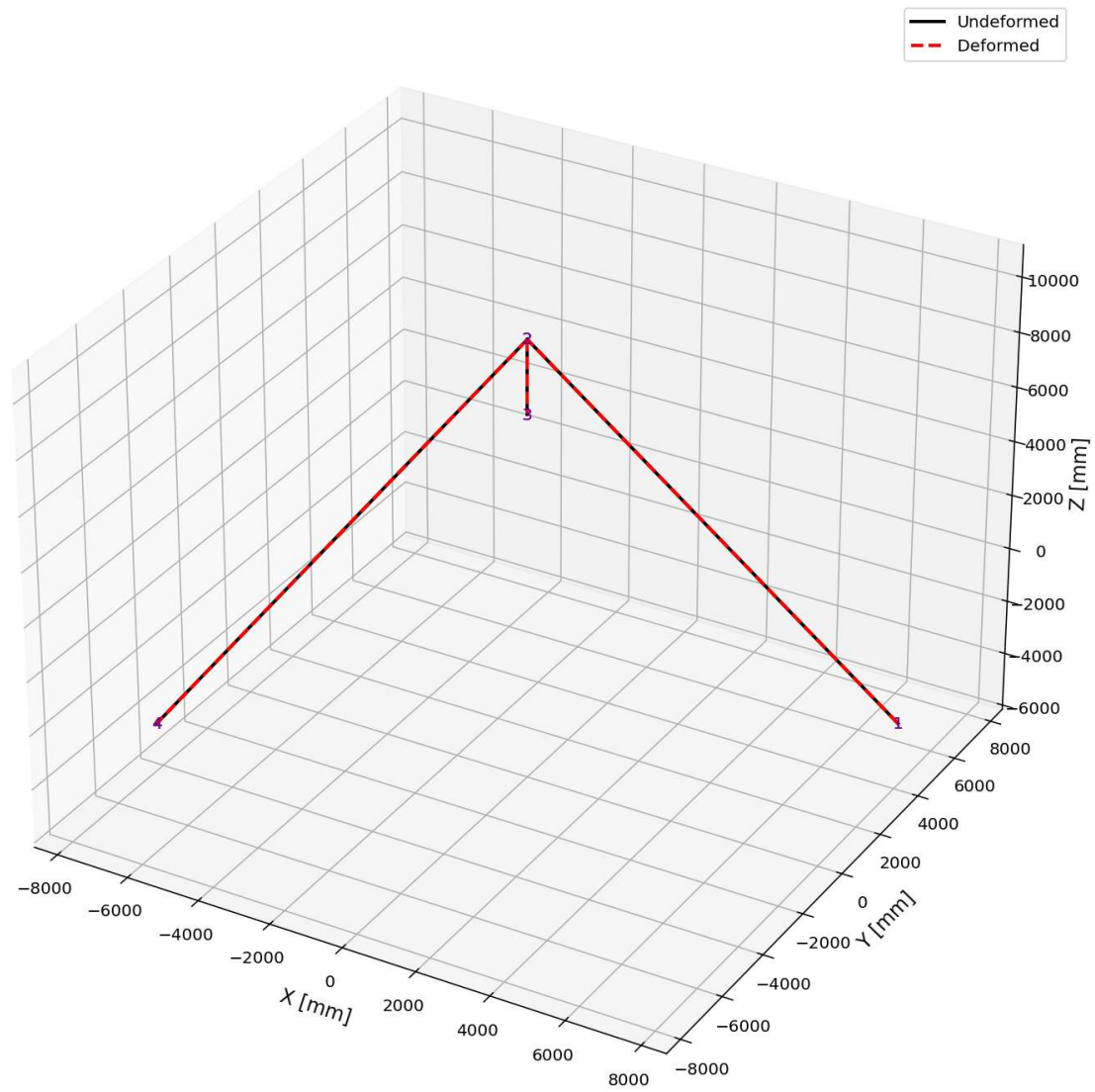


Node Displacements in Y Dir. vs Time for Truss Element During Dynamic External Time-dependent Loading Analysis





3D Truss: Undeformed vs Deformed (Scale = 1.00)



FREE-VIBRATION ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\EIGHT_ANALY...MENTS_8_ANA\P(t)_3D_TRUSS_THREE_ELEMENTS_8_ANA.py

P(t)_3D_TRUSS_THREE_ELEMENTS_8_ANA.py x PLOT_3D_TRUSS.py x

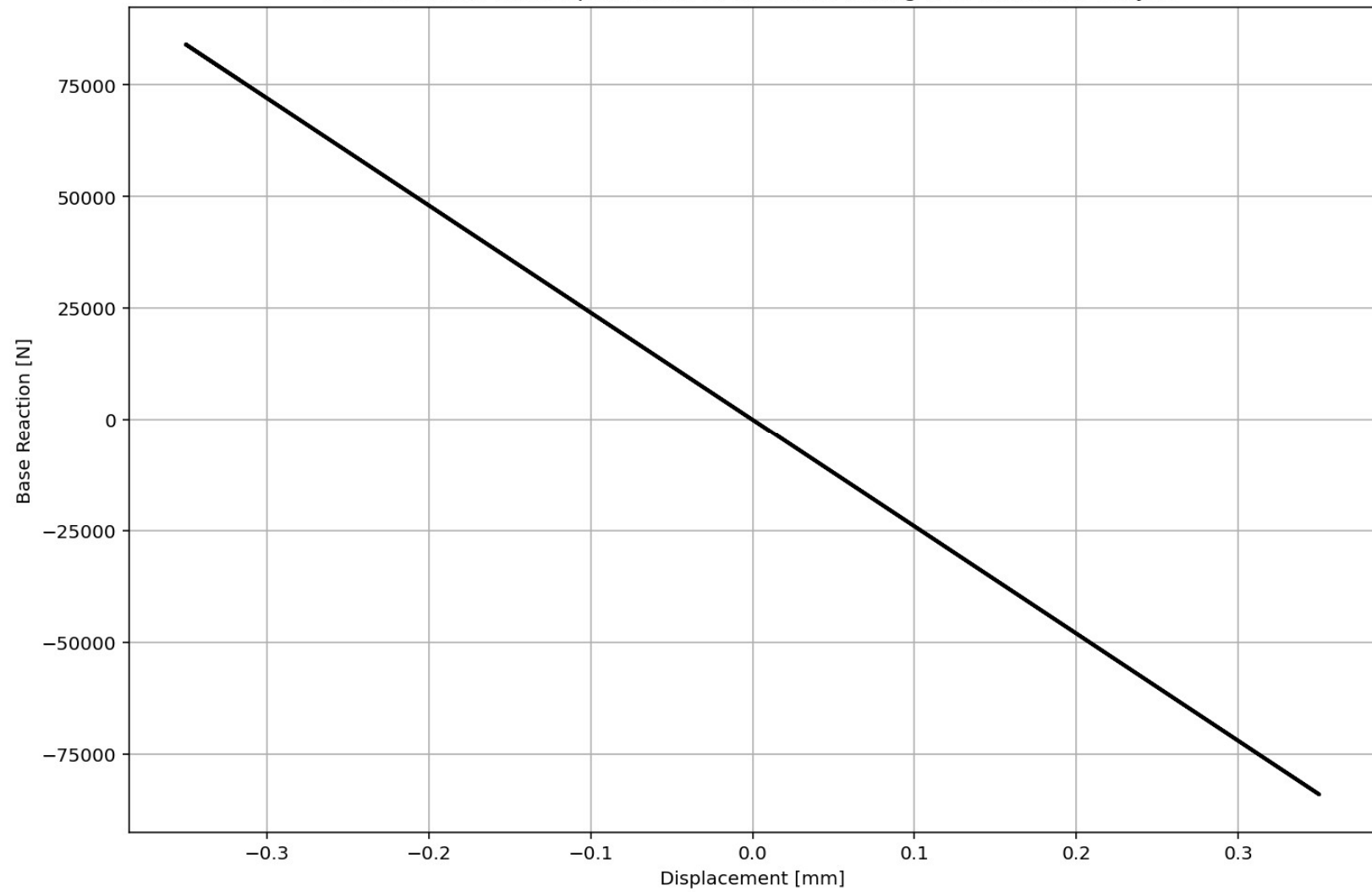
```
694
695 if ANAL TYPE == 'FREE-VIBRATION': # DYNAMIC TIME-HISTORY ANALYSIS
696     %% DEFINE PARAMETERS FOR FREE-VIBRATION ANALYSIS
697     u0 = -0.35 # [mm] Initial displacement
698     v0 = 0.15 # [mm/s] Initial velocity
699     a0 = 0.065 # [mm/s^2] Initial acceleration
700     IU = True # Free Vibration with Initial Displacement
701     IV = True # Free Vibration with Initial Velocity
702     IA = True # Free Vibration with Initial Acceleration
703     duration = 20.0 # [s] Analysis duration
704     dt = 0.001 # [s] Time step
705     DR = 0.03 # Damping Ratio
706
707     if IU == True:
708         # Define initial displacment
709         ops.setNodeDisp(center_node, 3, u0, '-commit') # INFO LINK: https://openseespy
710     if IV == True:
711         # Define initial velocity
712         ops.setNodeVel(center_node, 3, v0, '-commit') # INFO LINK: https://openseespy
713     if IA == True:
714         # Define initial acceleration
715         ops.setNodeAccel(center_node, 3, a0, '-commit') # INFO LINK: https://openseespy
716
717     ops.constraints('Plain')
718     ops.numberer('Plain')
719     ops.system('BandGeneral')
720     ops.test('NormDispIncr', MAX_TOLERANCE, MAX_ITERATIONS) # INFO LINK: https://openseespydoc.readthedocs.io/en/latest/algorithm.html
721     #ops.integrator('CentralDifference') # JUST FOR LINEAR ANALYSIS - INFO LINK: https://openseespydoc.readthedocs.io/en/latest/algorithm.html
722     alpha=0.5; beta=0.25;
723     ops.integrator('Newmark', alpha, beta) # INFO LINK: https://openseespydoc.readthedocs.io/en/latest/algorithm.html
724     #alpha=2/3; gamma=1.5-alpha; gamma=1.5-alpha; beta=(2-alpha)**2/4;
725     #ops.integrator('HHT', alpha, gamma, beta) # INFO LINK: https://openseespydoc.readthedocs.io/en/latest/algorithm.html
726     ops.algorithm('Newton') # INFO LINK: https://openseespydoc.readthedocs.io/en/latest/algorithm.html
727     ops.analysis('Transient') # INFO LINK: https://openseespydoc.readthedocs.io/en/latest/algorithm.html
```

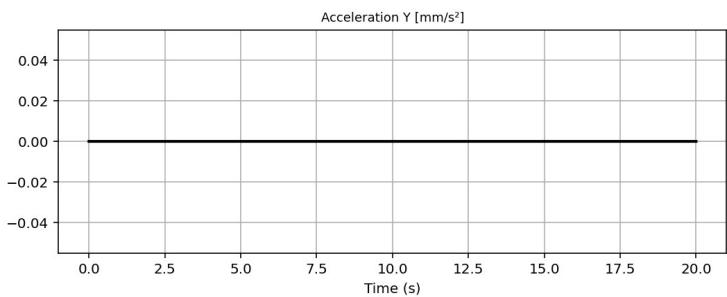
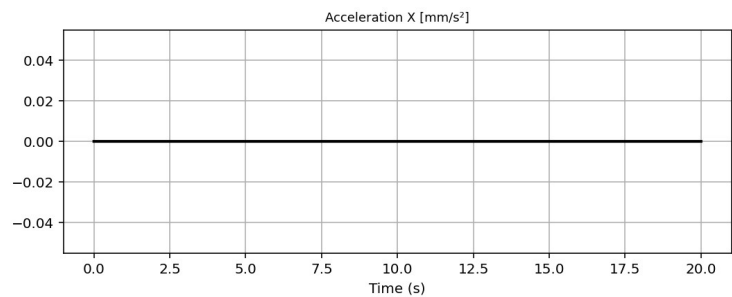
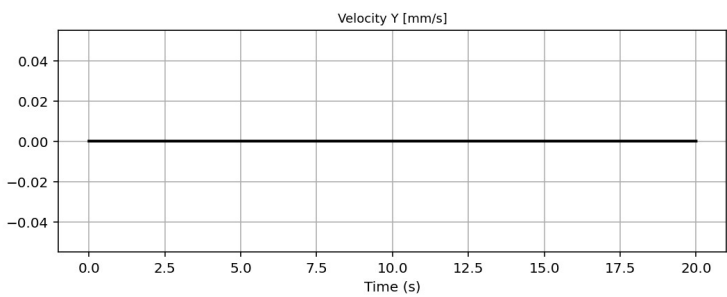
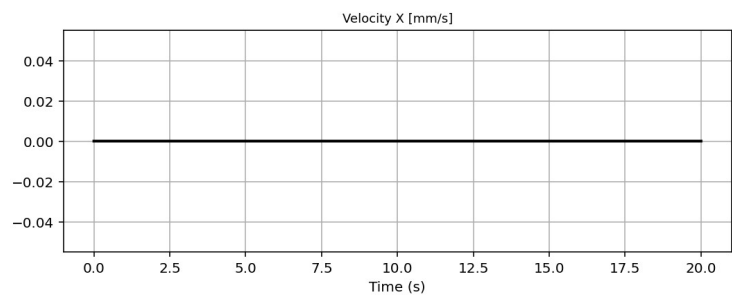
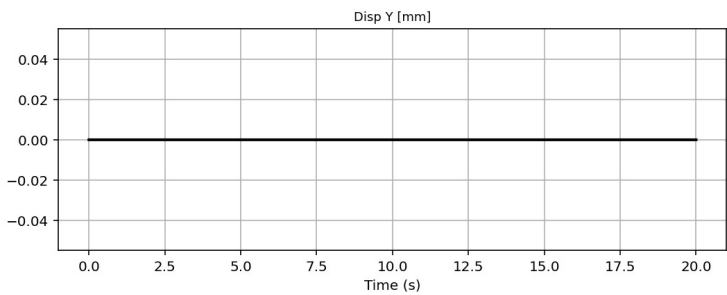
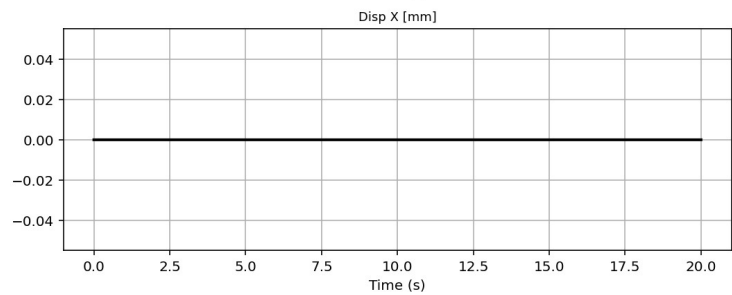
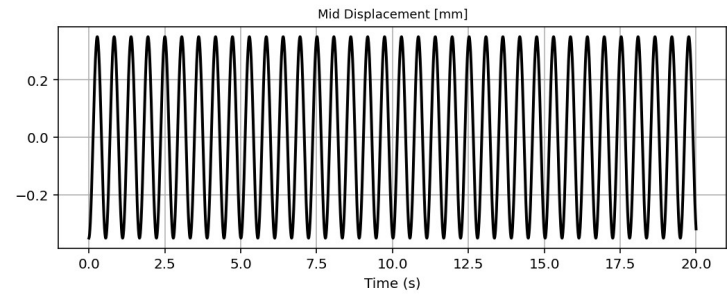
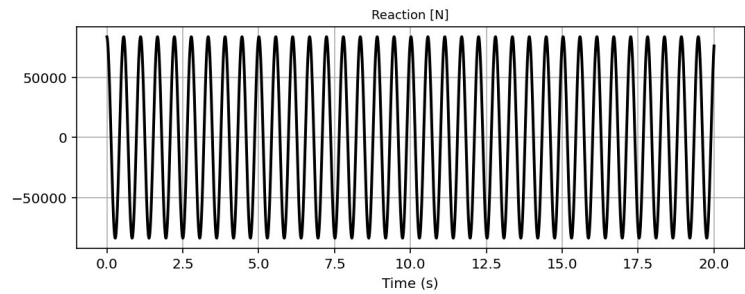
Base Reaction and Displacement of Structure During Free-vibration Analysis

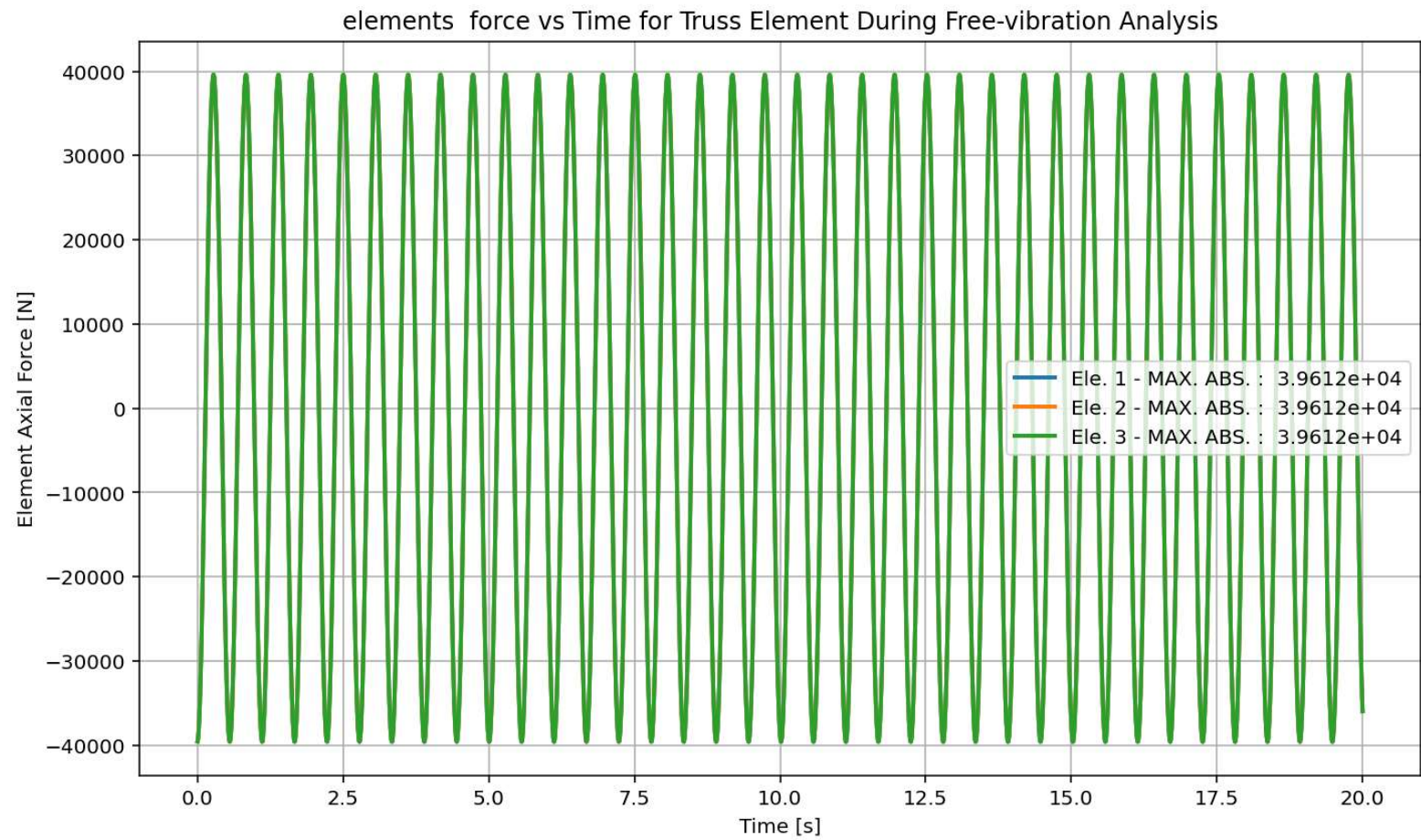
[Python Console] Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 302, Col 27 UTF-8 CRLF RW Mem 56%

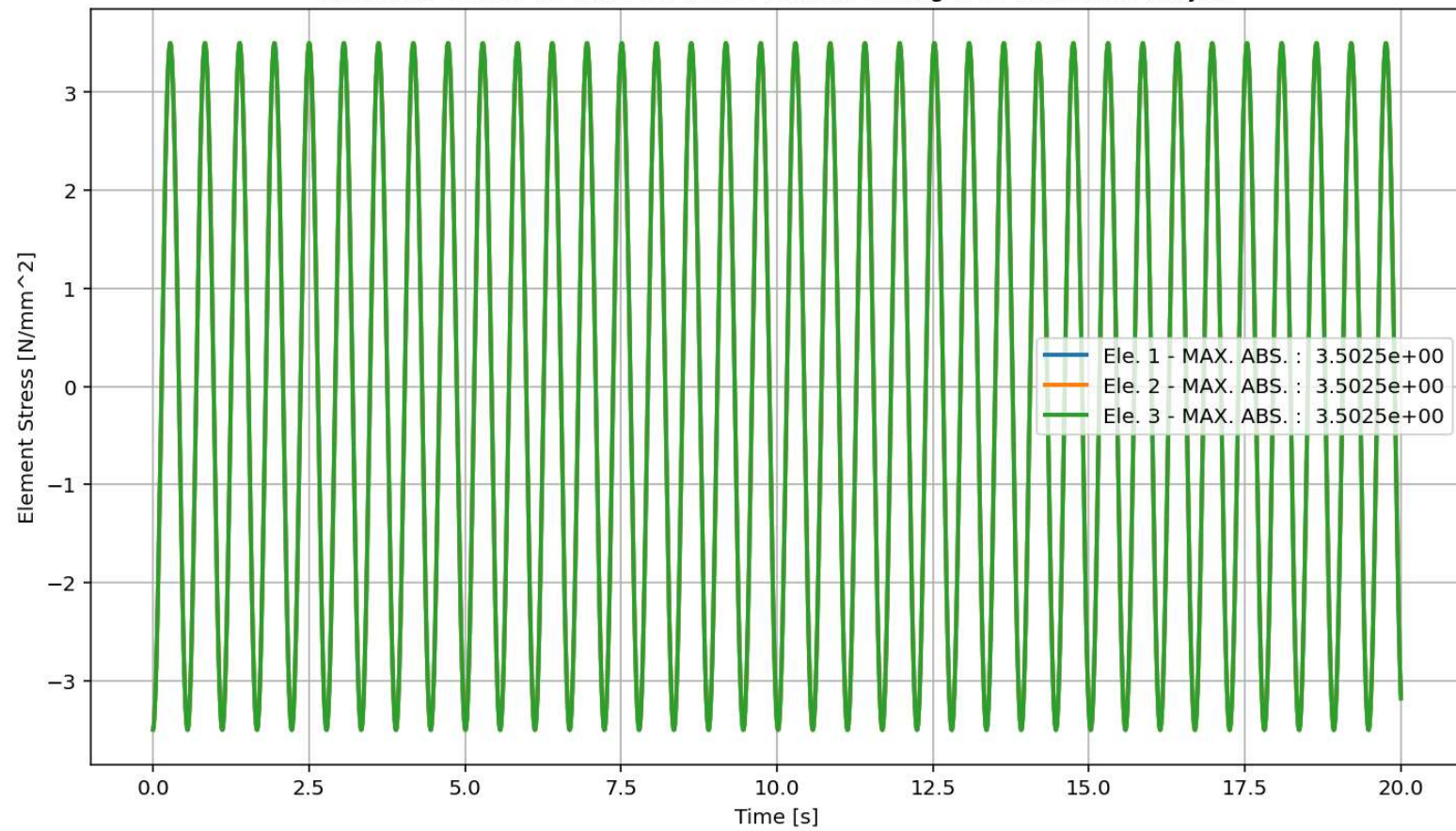
Base Reaction and Dispalcement of Structure During Free-vibration Analysis

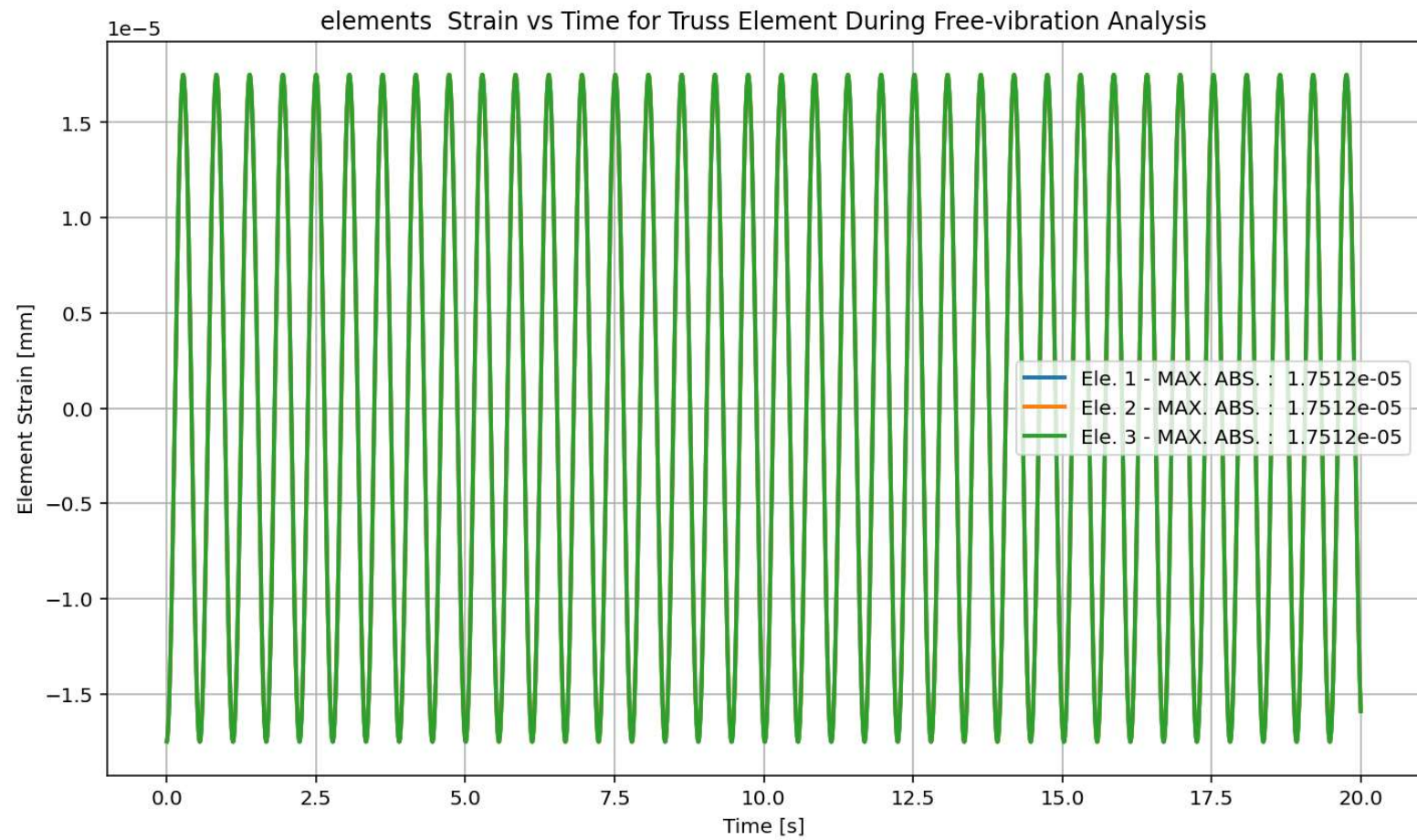


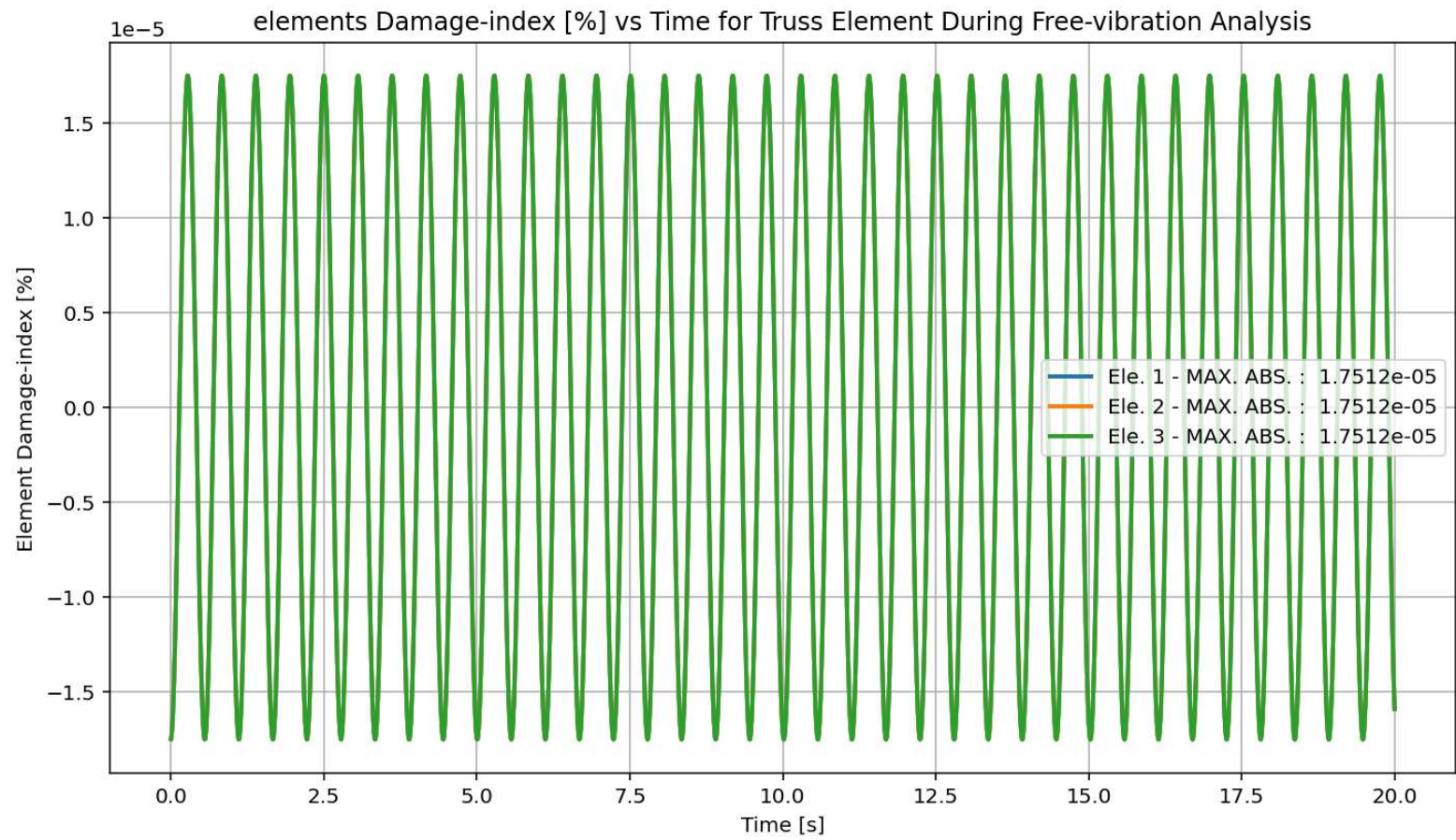


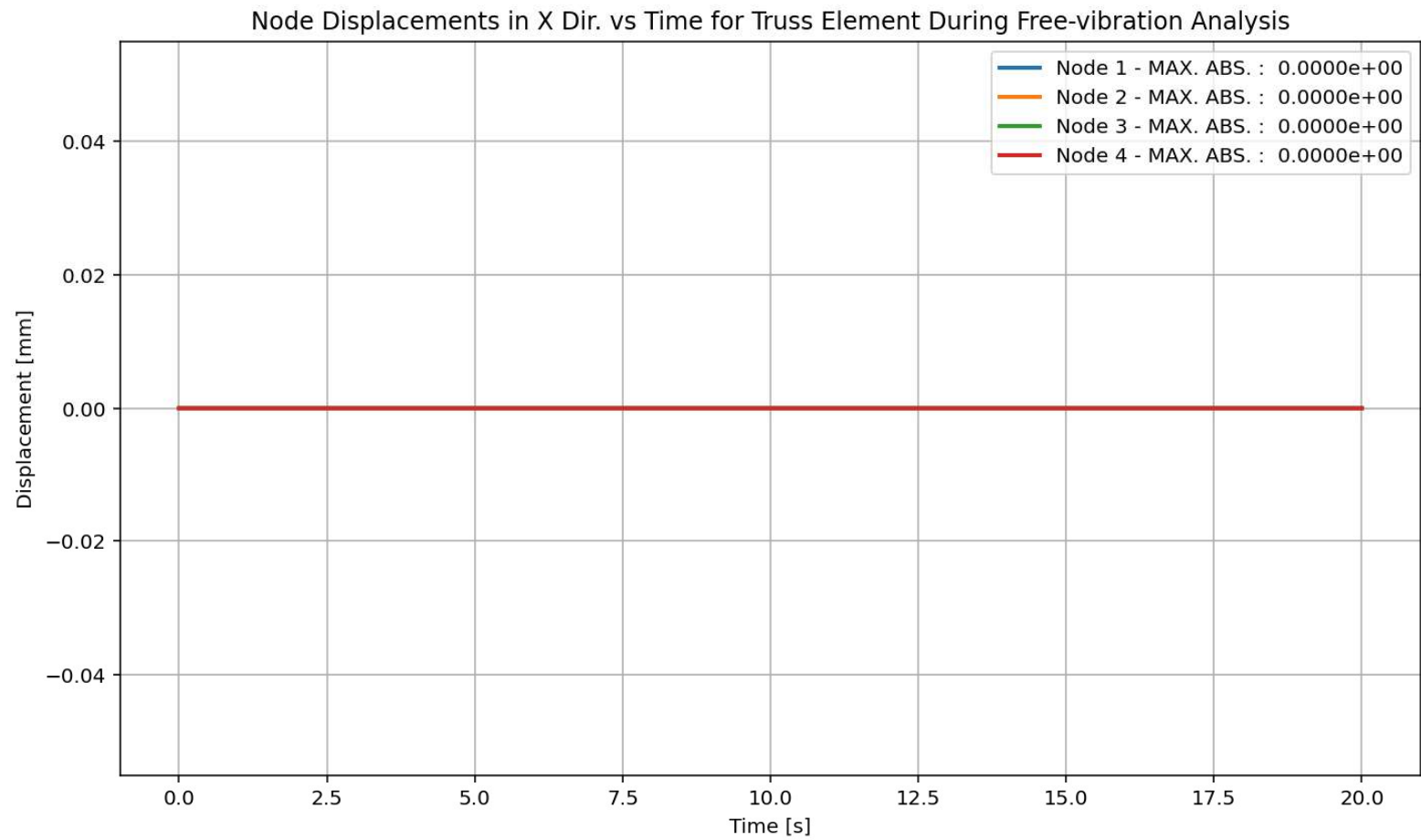


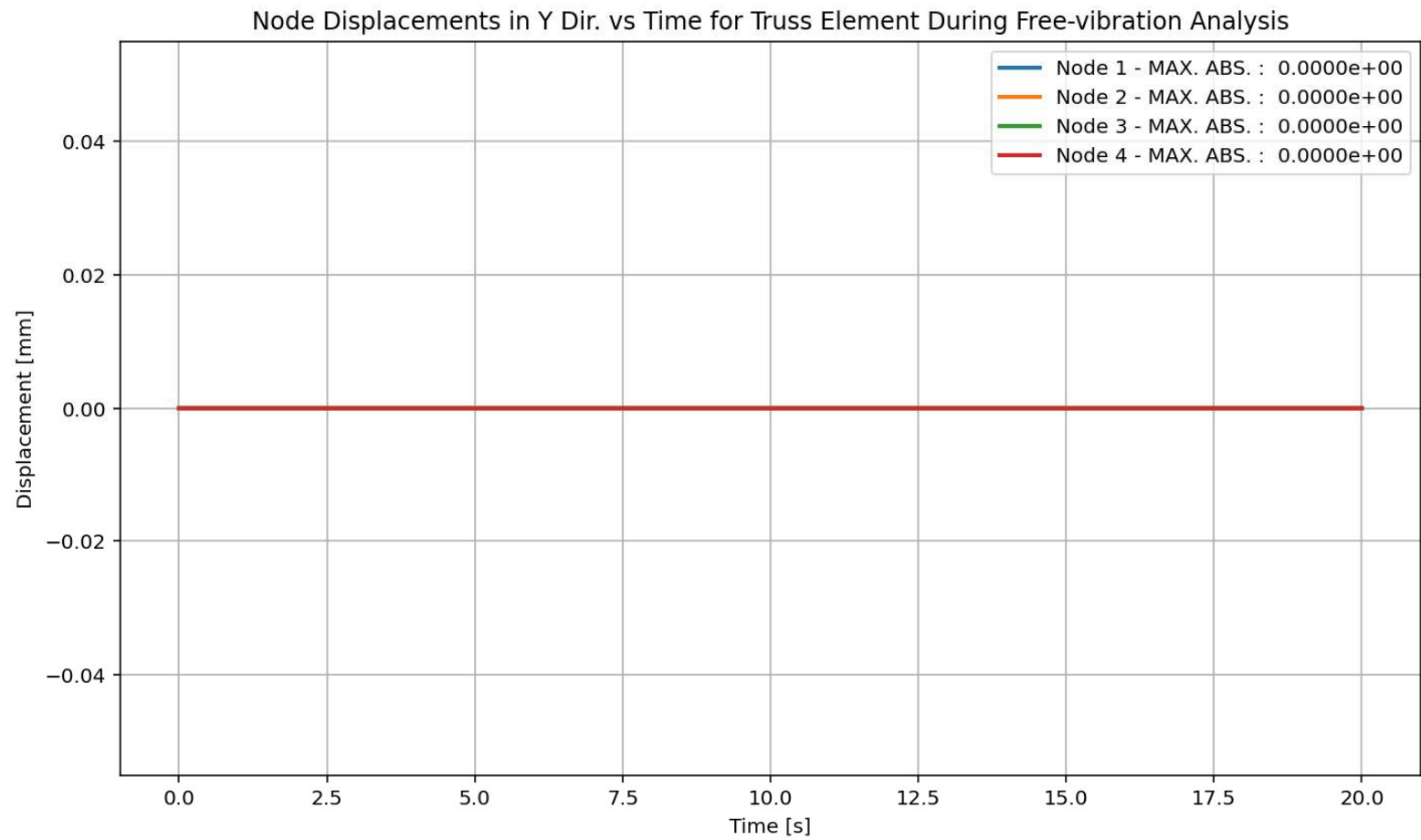
elements Stress vs Time for Truss Element During Free-vibration Analysis

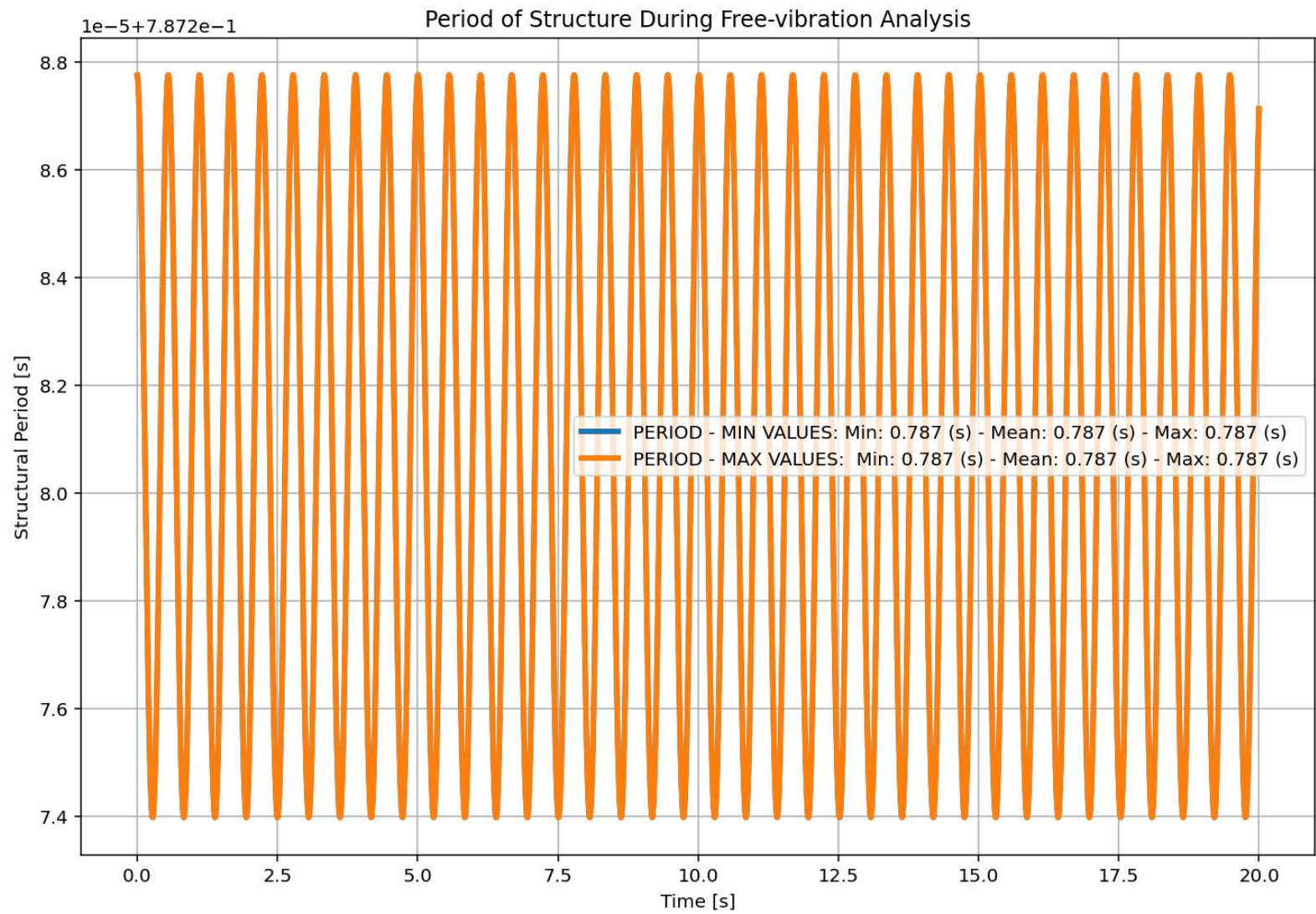




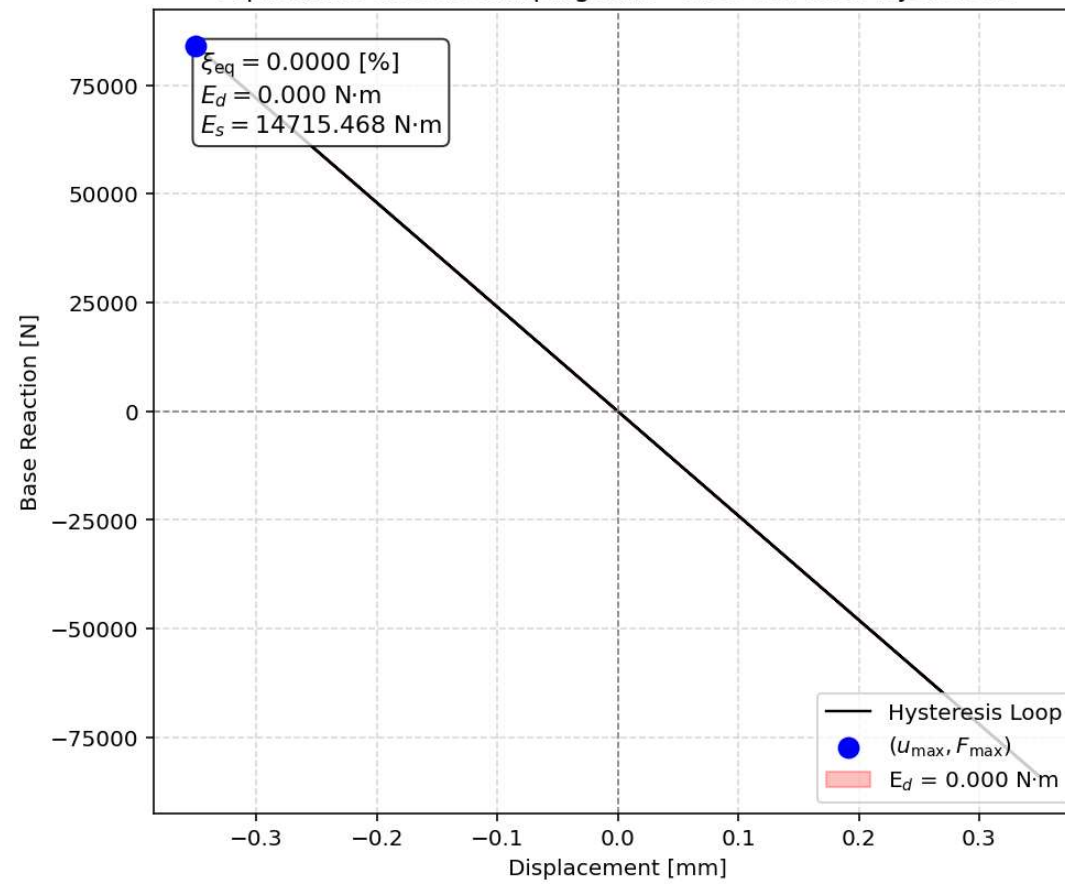




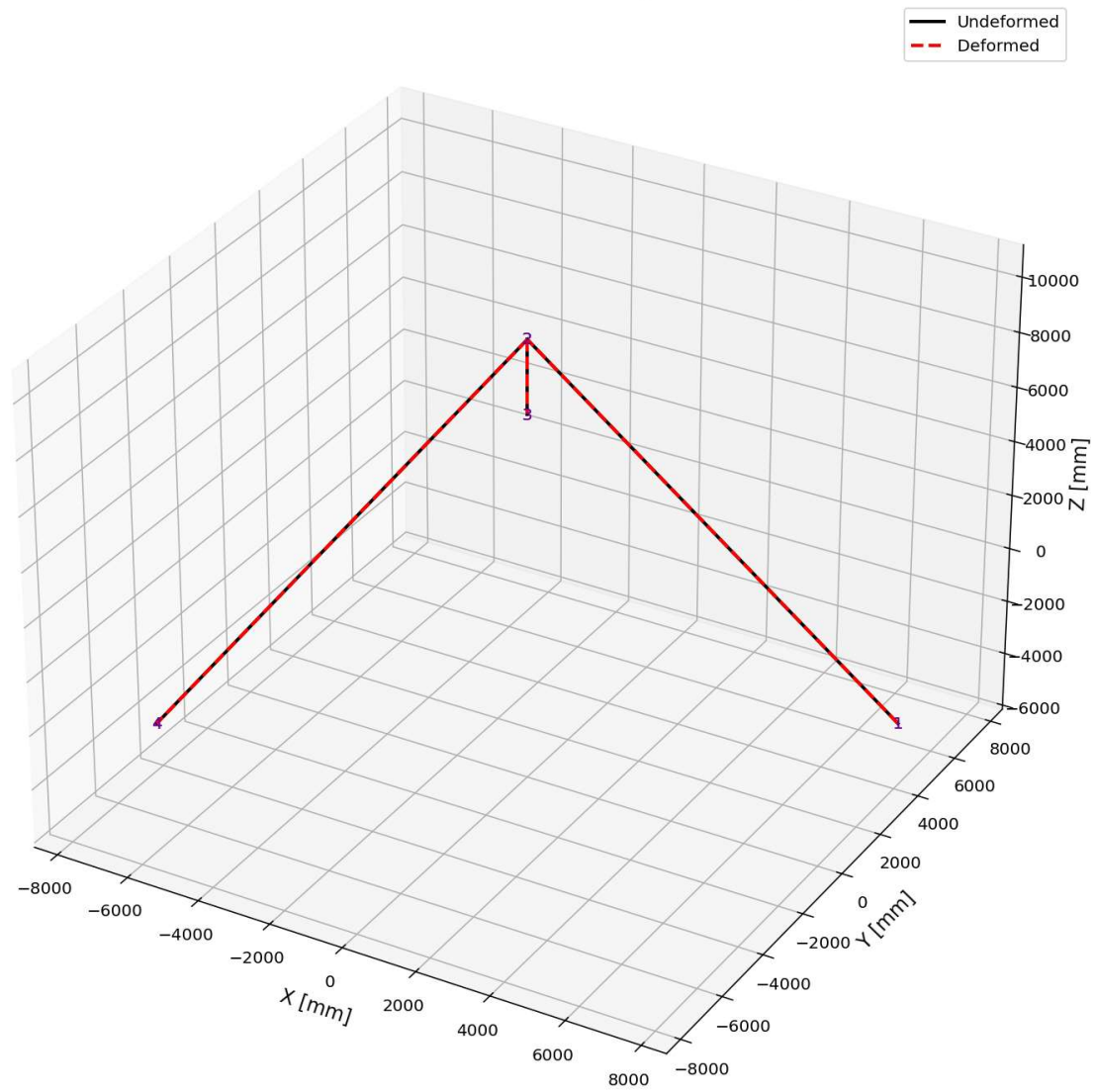




Equivalent viscous damping ratio - Free-vibration Hysteresis



3D Truss: Undeformed vs Deformed (Scale = 1.00)



SEISMIC ANALYSIS

File Edit Search Source Run Debug Consoles Projects Tools View Help

...DISSIPATION_CAPACITY_INDEX\34_3D_TRUSS_THREE_ELEMENTS_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MENTS_8_ANA\p(t)_3D_TRUSS_THREE_ELEMENTS_8_ANA.py

P(t)_3D_TRUSS_THREE_ELEMENTS_8_ANA.py x PLOT_3D_TRUSS.py x

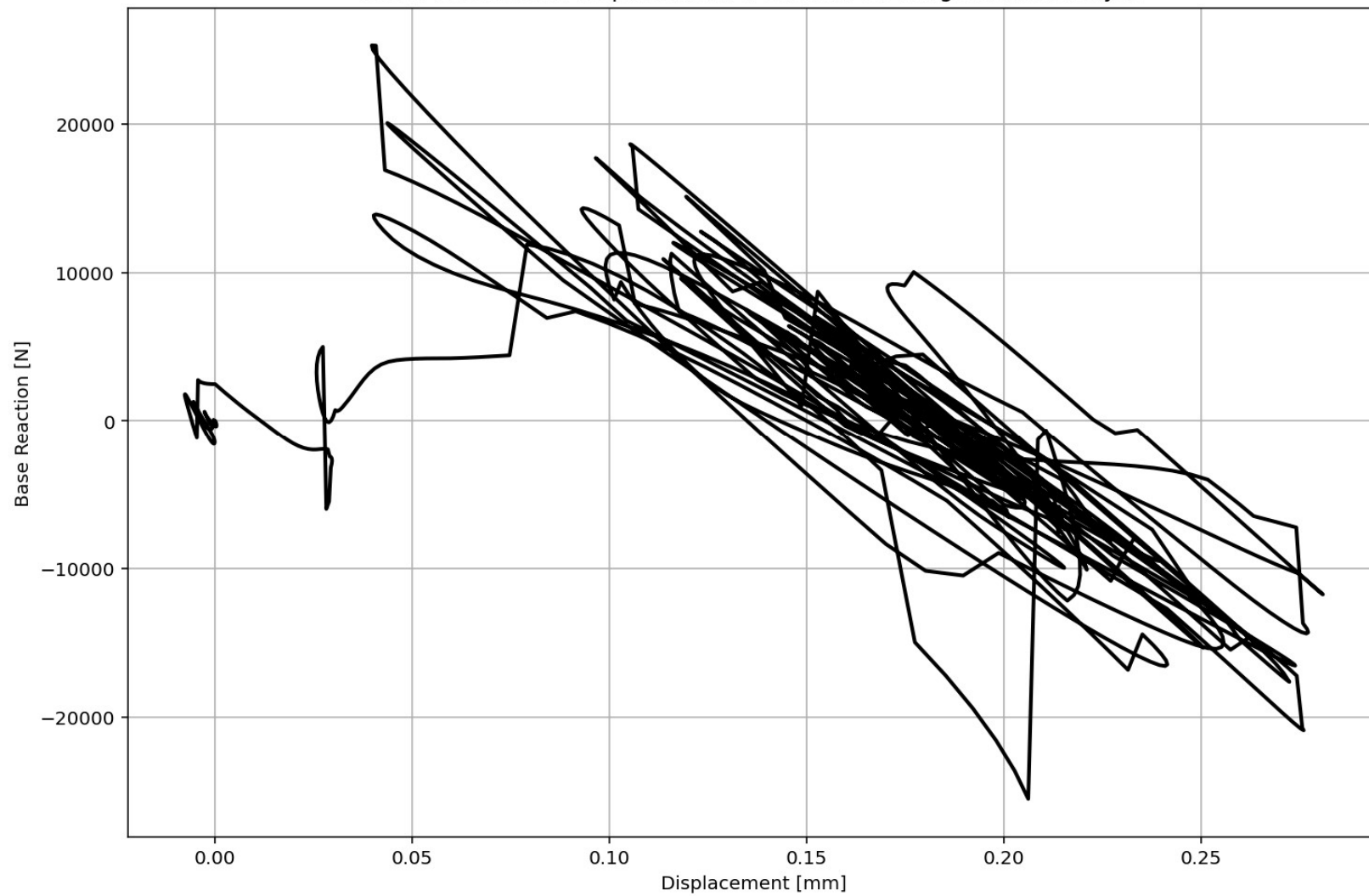
```
793 if ANAL_TYPE == 'SEISMIC': # DYNAMIC TIME-HISTORY ANALYSIS
794     %% DEFINE PARAMETERS FOR SEISMIC ANALYSIS
795     duration = 20.0 # [s] Analysis duration
796     dt = 0.01 # [s] Time step
797     GMfact = 9810 # [mm/s^2] standard acceleration of gravity or st
798     SSF_X = 10.50 # Seismic Acceleration Scale Factor in X Direct
799     SSF_Y = 10.50 # Seismic Acceleration Scale Factor in Y Direct
800     iv0_X = 0.0005 # [mm/s] Initial velocity applied to the node
801     iv0_Y = 0.0005 # [mm/s] Initial velocity applied to the node
802     st_iv0 = 0.0 # [s] Initial velocity applied starting time
803     SEI = 'Y' # Seismic Direction
804     DR = 0.03 # Damping ratio
805
806 # Define time series for input motion (Acceleration time history)
807 if SEI == 'X':
808     SEISMIC_TAG_01 = 100
809     ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath', 'Ground_Acceler
810     # Define load patterns
811     # pattern UniformExcitation $patternTag $dof -accel $tsTag <-vel0 $vel0> <-fac
812     ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', SEISMIC_TAG_01,
813 if SEI == 'Y':
814     SEISMIC_TAG_02 = 200
815     ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath', 'Ground_Acceler
816     ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', SEISMIC_TAG_02,
817 if SEI == 'XY':
818     SEISMIC_TAG_01 = 100
819     ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath', 'Ground_Acceler
820     # Define load patterns
821     # pattern UniformExcitation $patternTag $dof -accel $tsTag <-vel0 $vel0> <-fac
822     ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', SEISMIC_TAG_01,
823     SEISMIC_TAG_02 = 200
824     ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath', 'Ground_Acceler
825     ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', SEISMIC_TAG_02,
826     print('Seismic Defined Done.')
```

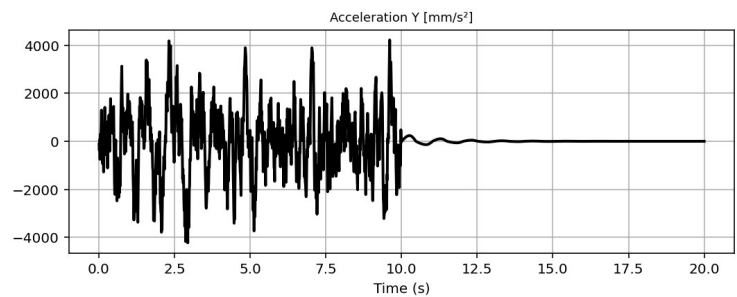
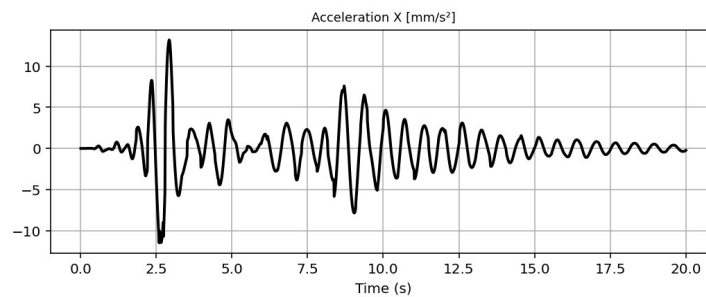
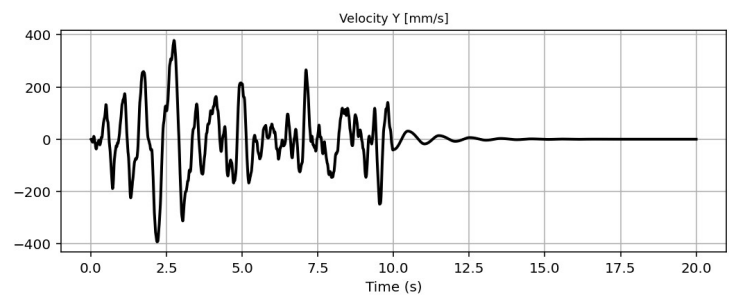
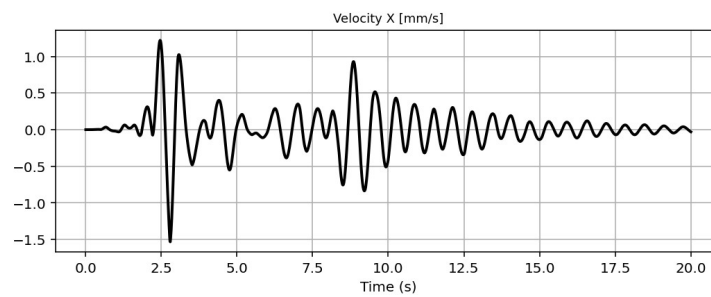
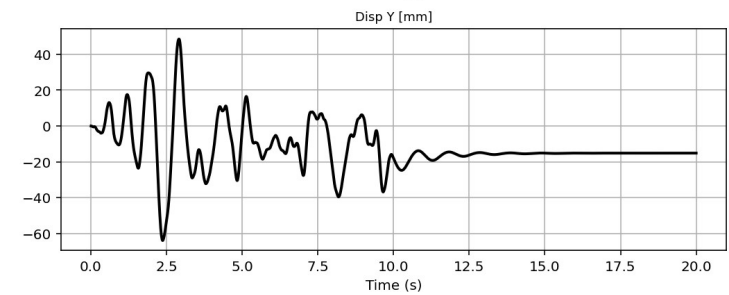
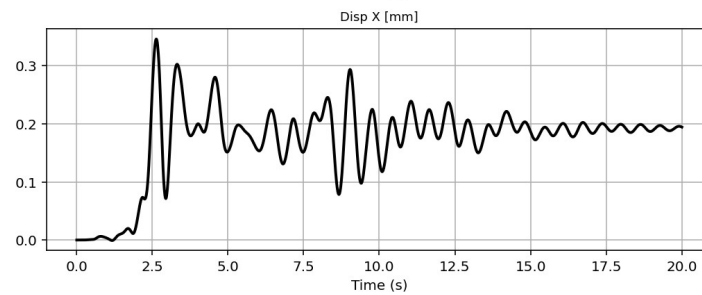
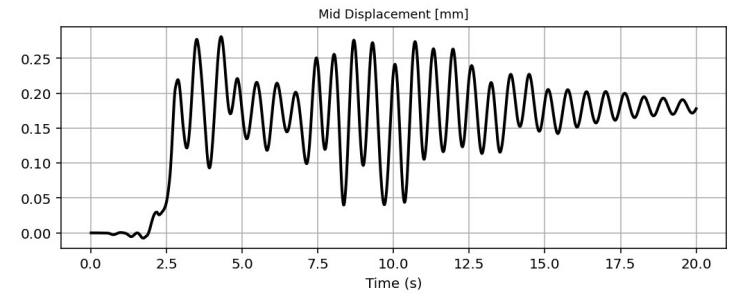
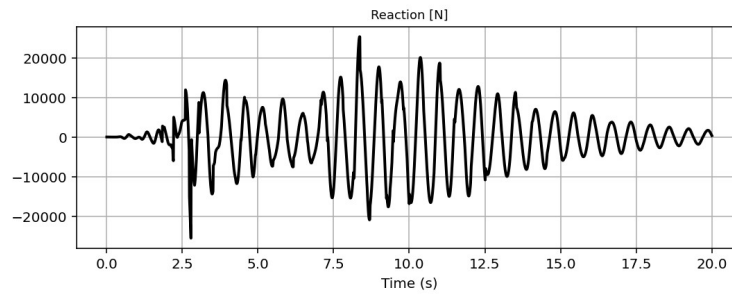
Base Reaction and Displacement of Structure During Seismic Analysis

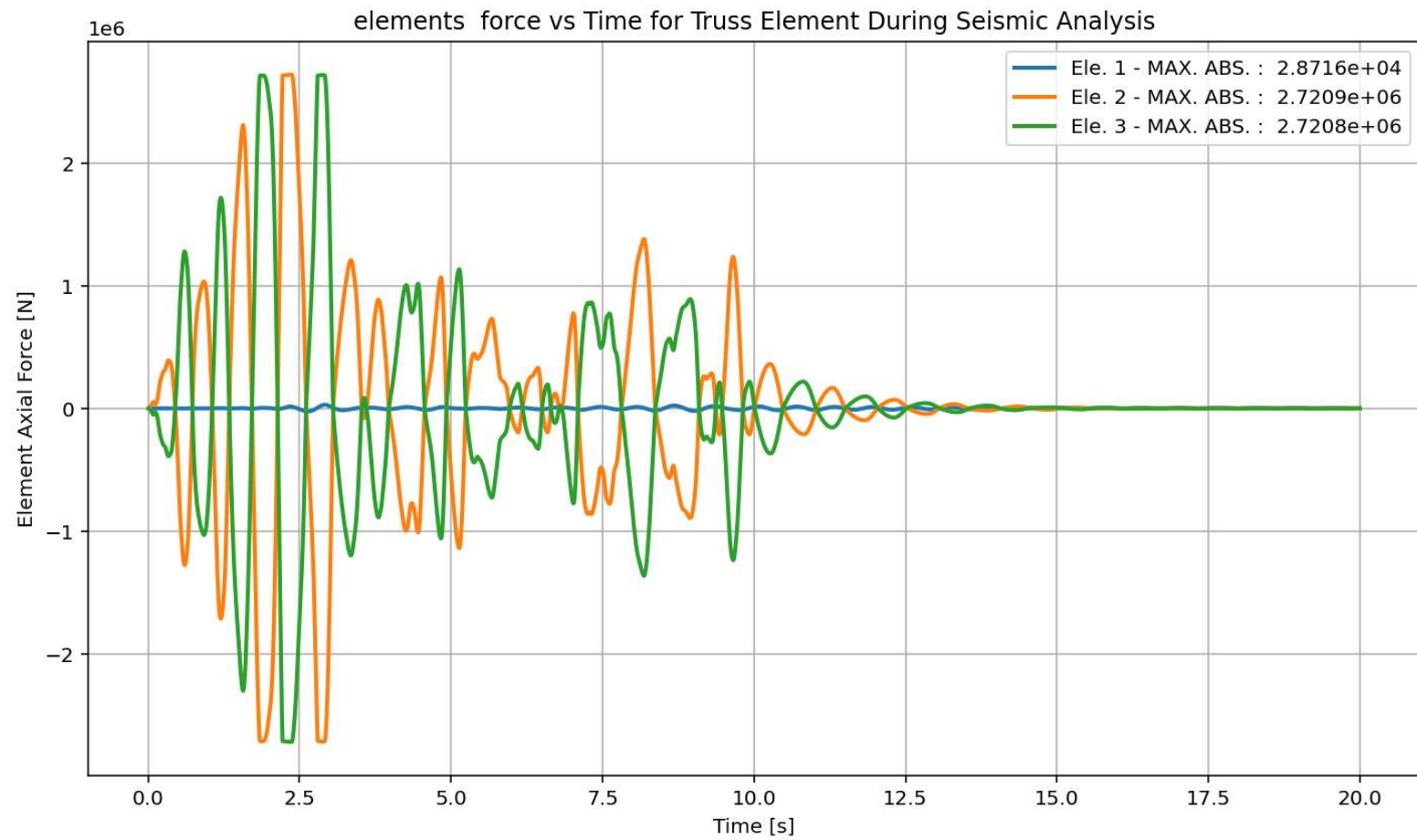
[Python Console] Files Help Variable Explorer Debugger Plots History

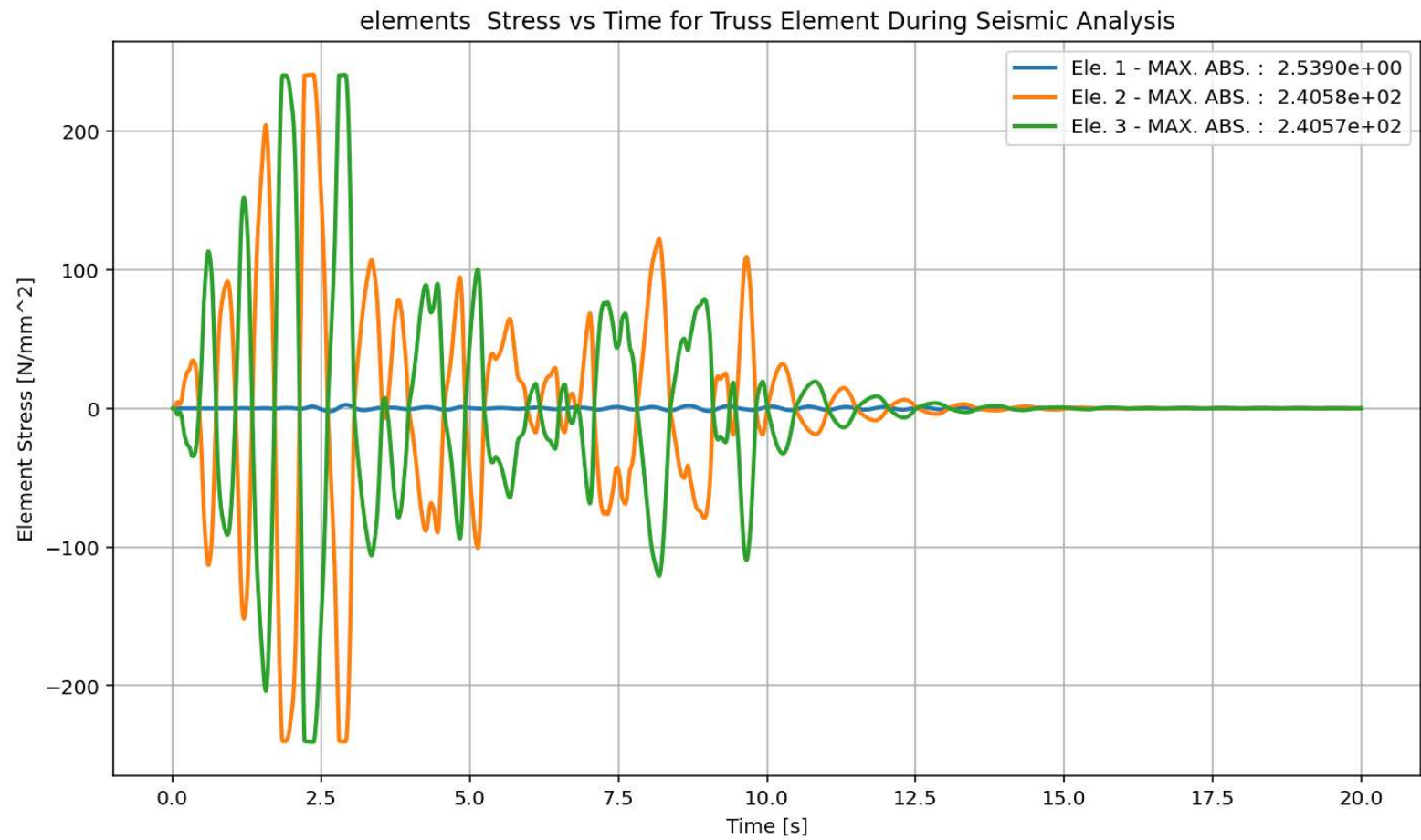
Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 1200, Col 26 UTF-8 CRLF RW Mem 36%

Base Reaction and Displacement of Structure During Seismic Analysis





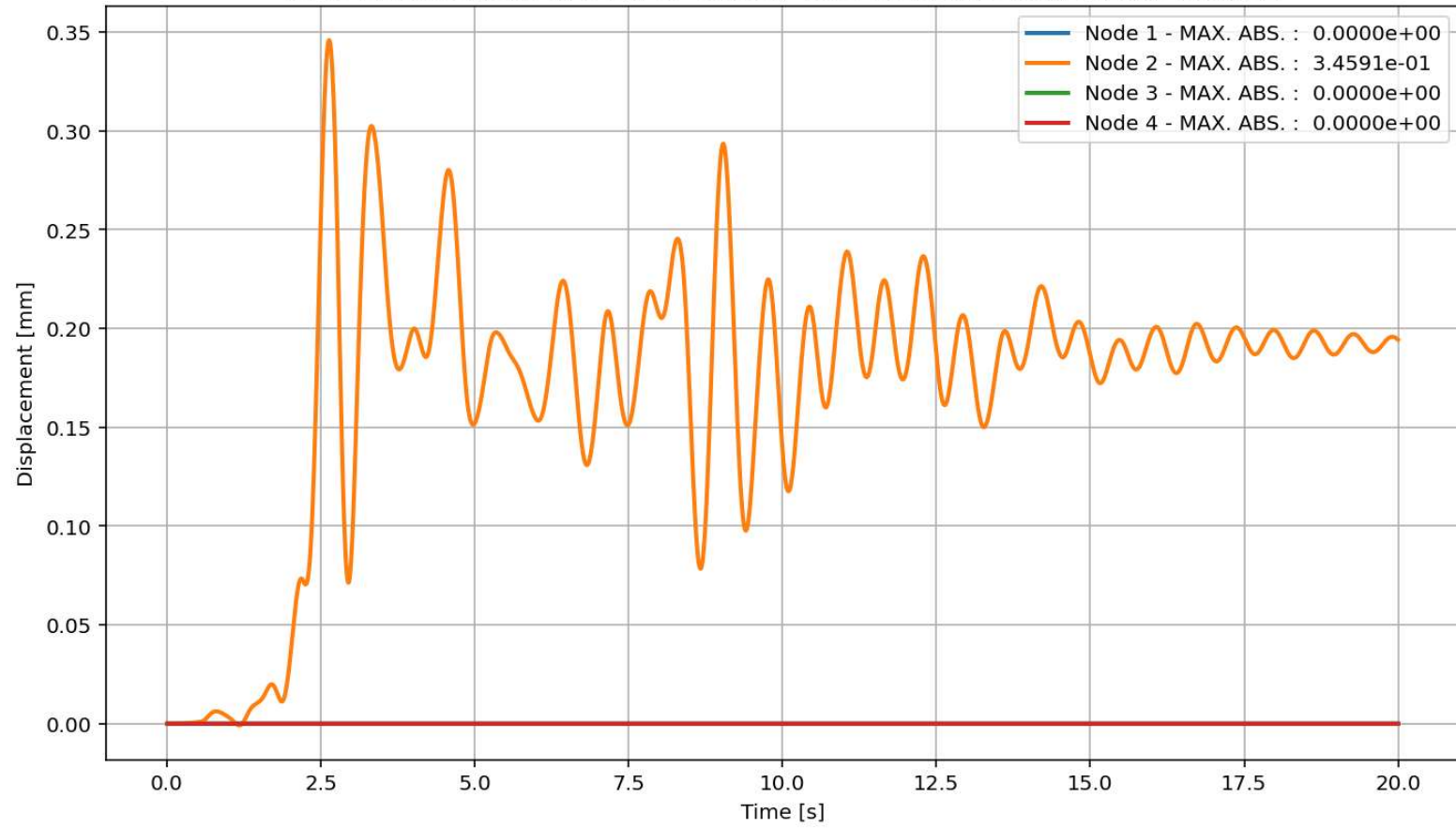




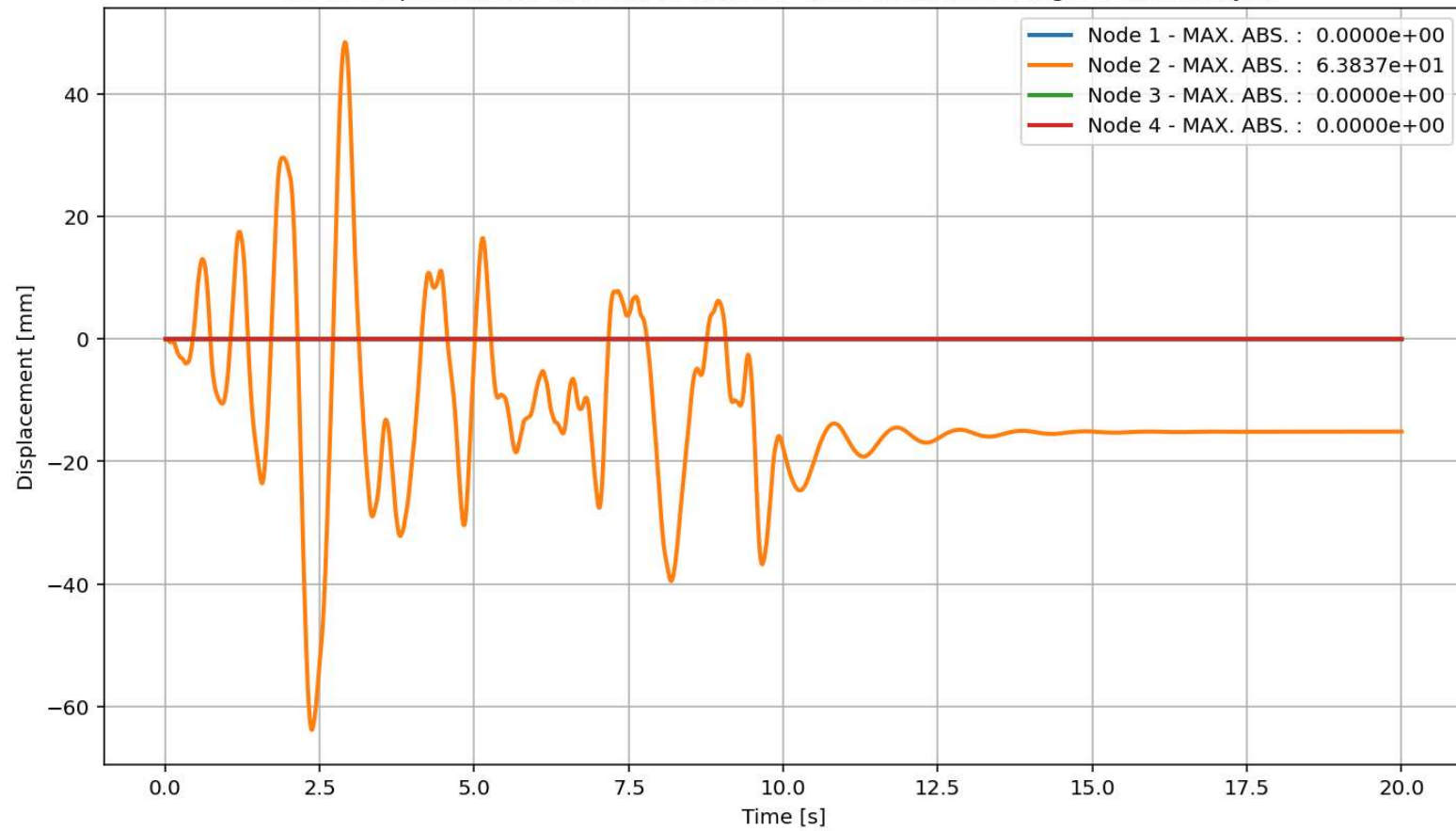




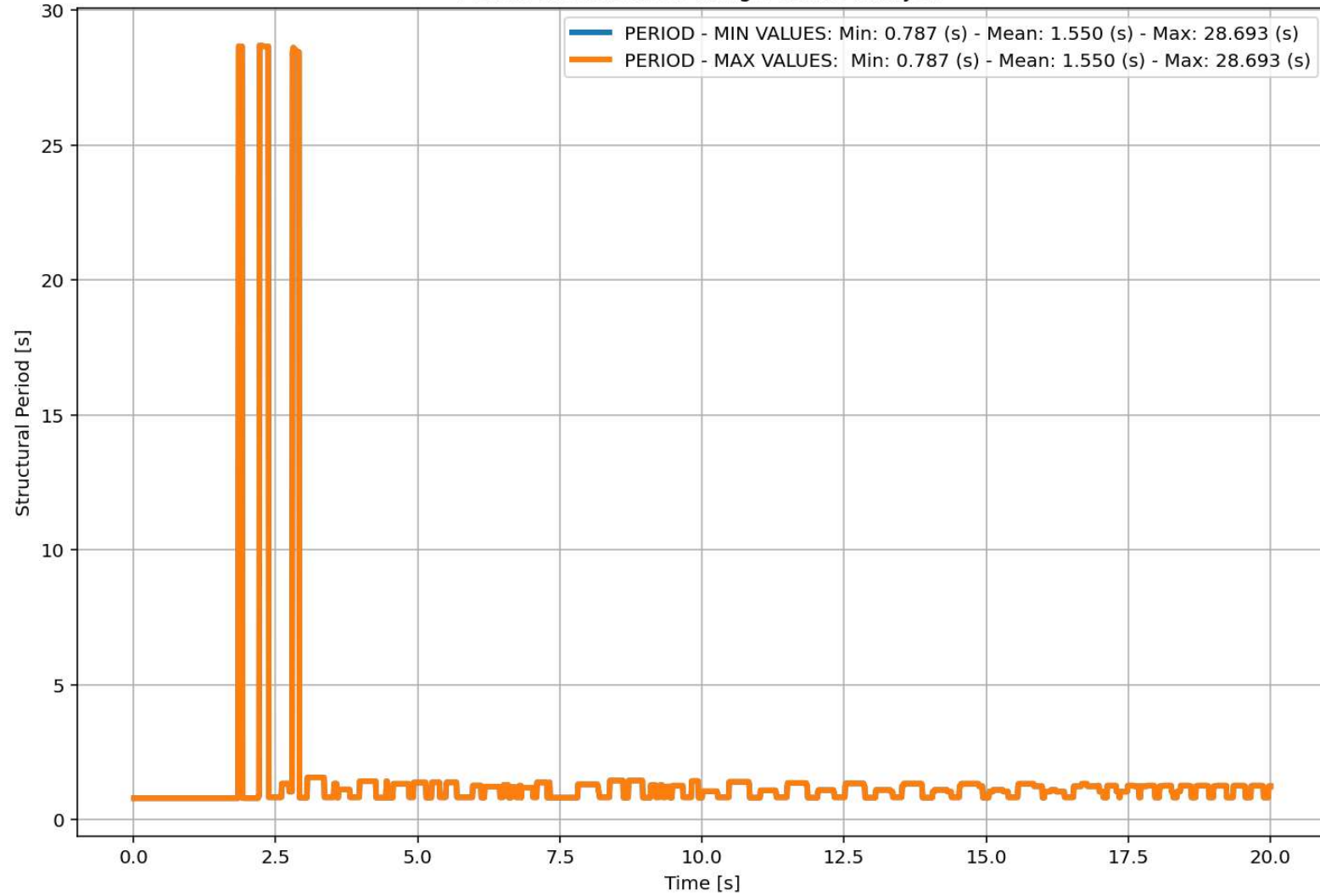
Node Displacements in X Dir. vs Time for Truss Element During Seismic Analysis



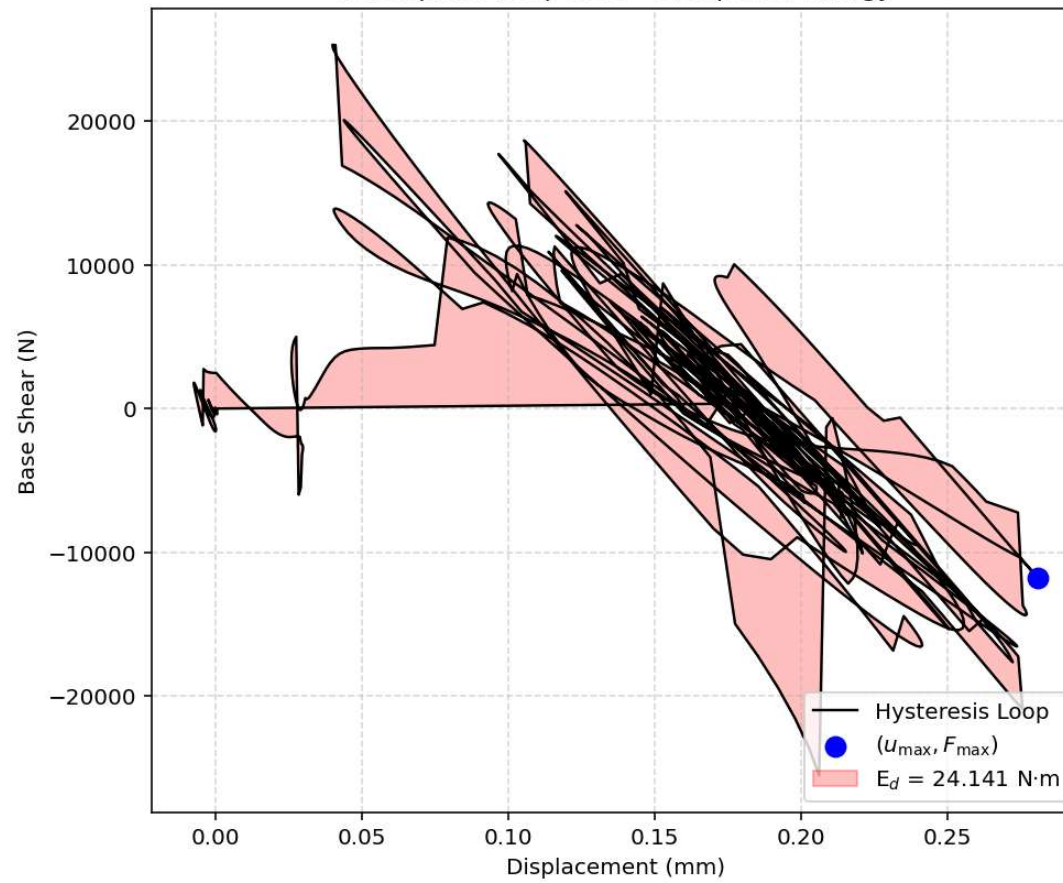
Node Displacements in Y Dir. vs Time for Truss Element During Seismic Analysis



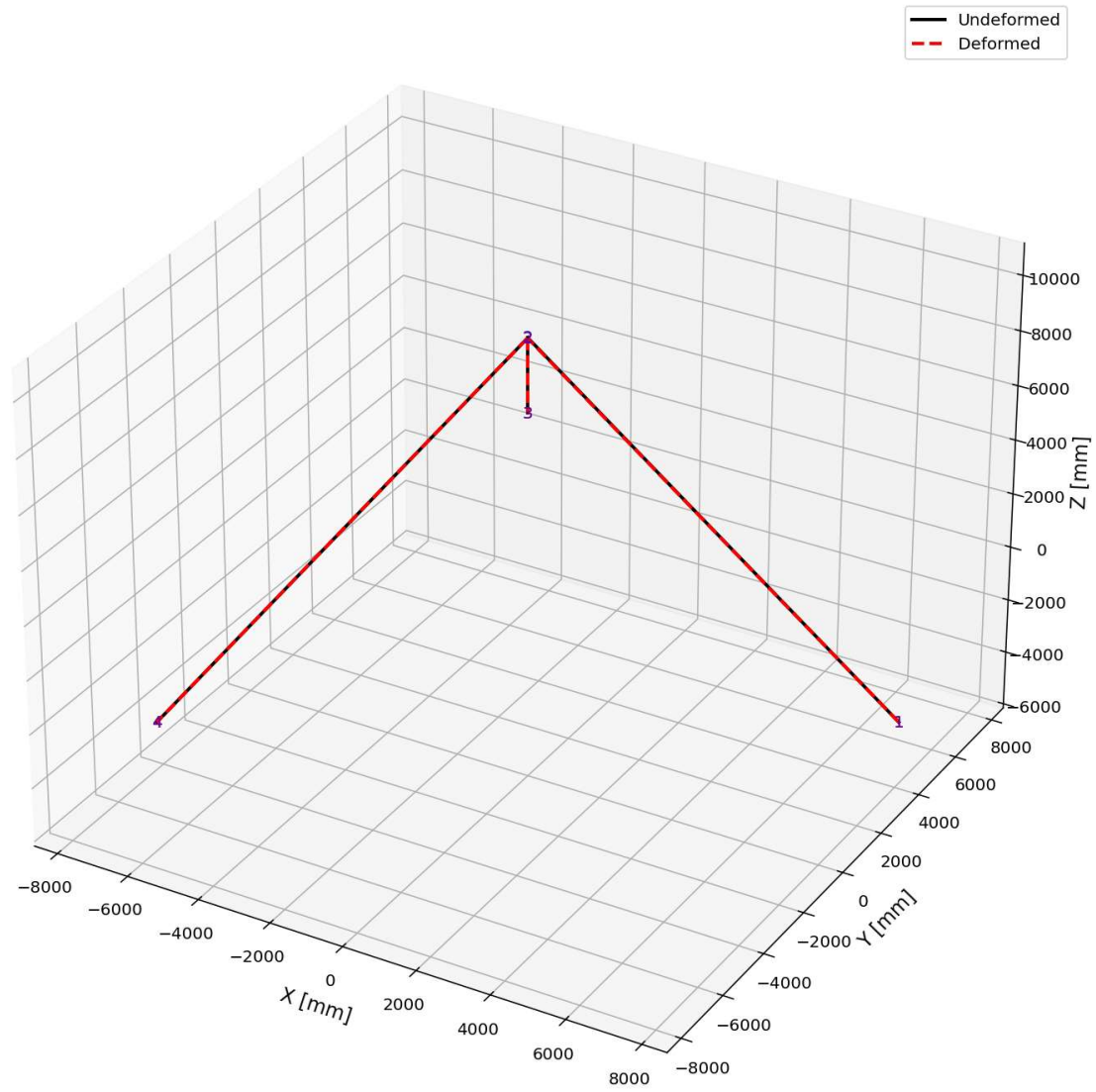
Period of Structure During Seismic Analysis

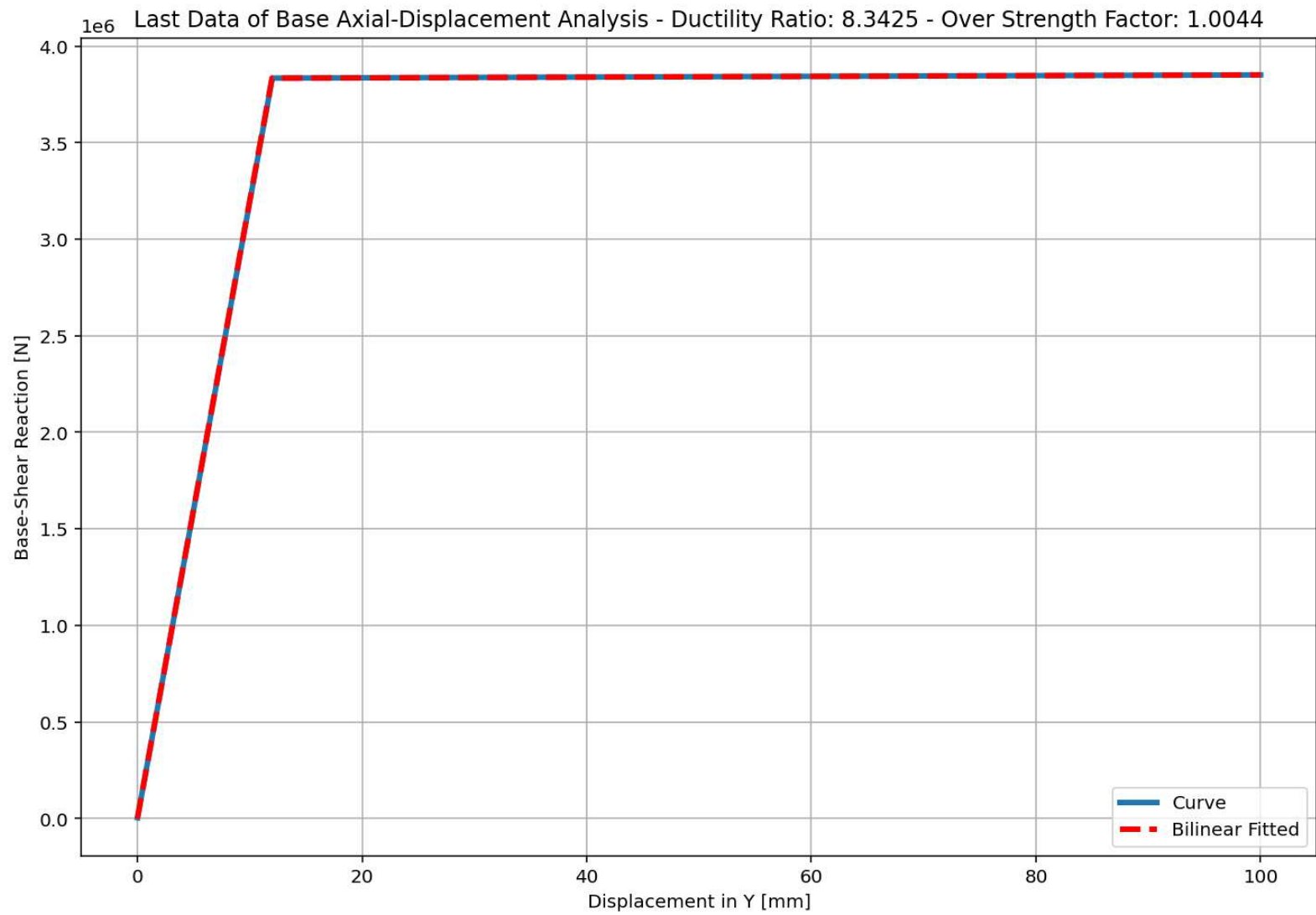


Earthquake Response - Dissipated Energy



3D Truss: Undeformed vs Deformed (Scale = 1.00)





+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]
[2.39648372e+01 5.74910463e+06]
[1.00000000e+02 5.76217409e+06]]

+=====+

+-----+

Structure Elastic Stiffness : 239897.50

Structure Plastic Stiffness : 57621.74

Structure Tangent Stiffness : 171.89

Structure Ductility Ratio : 4.17

Structure Over Strength Factor: 1.00

+-----+

Over Strength Coefficient (Ω_0): 1.0023

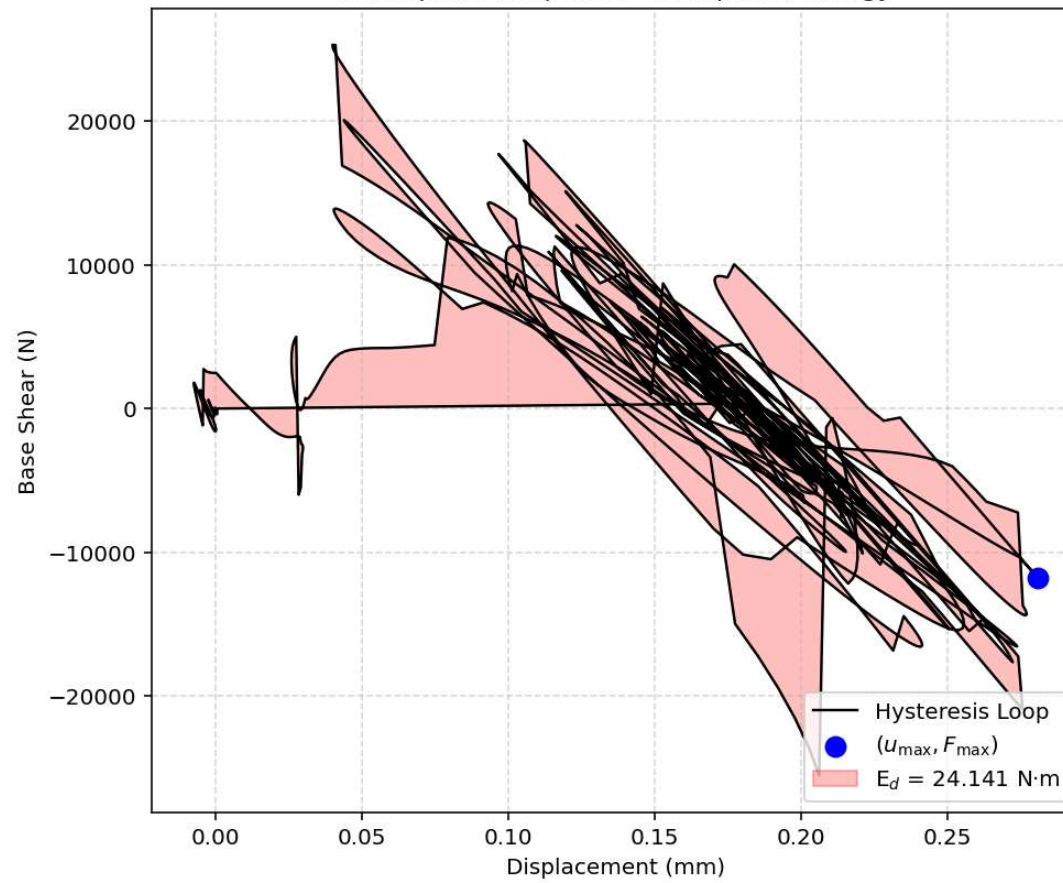
Displacement Ductility Ratio (μ): 4.1728

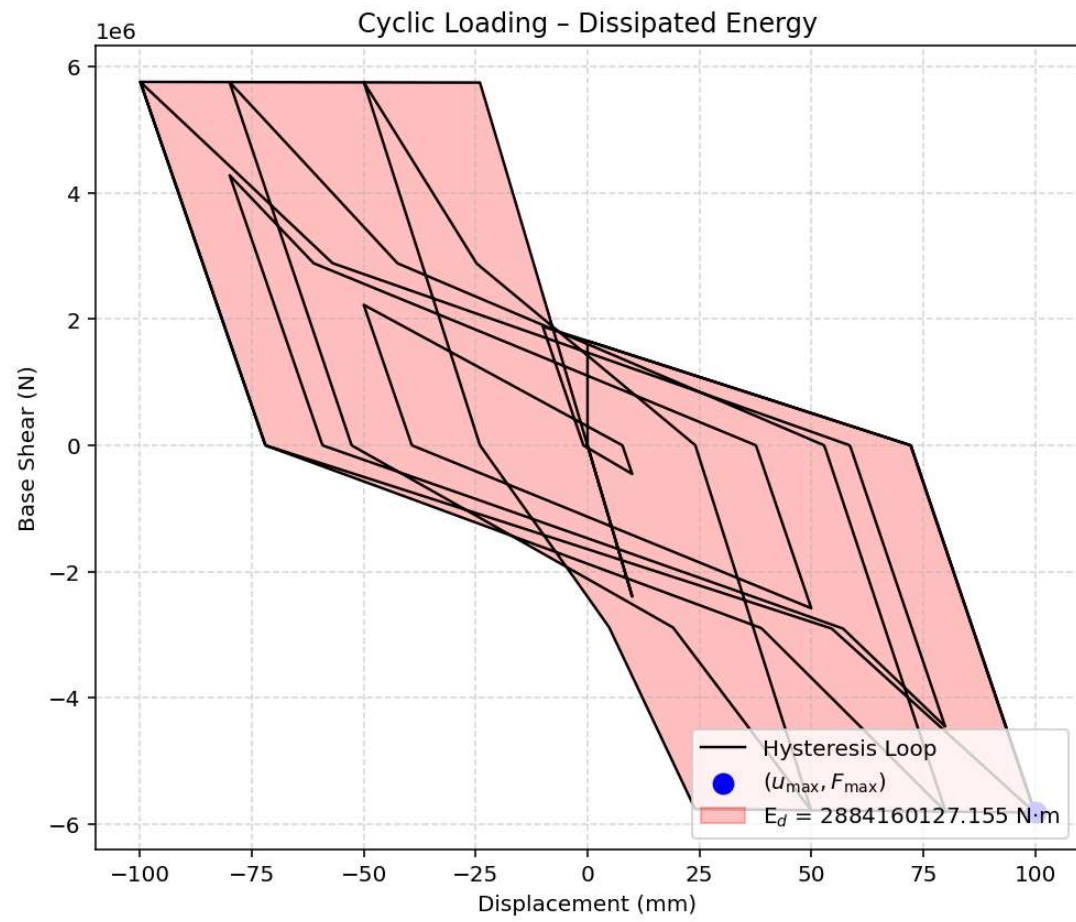
Ductility Coefficient (R_μ): 4.1728

Structural Behavior Coefficient (R): 4.1823

Structural Ductility Damage Index in Y Direction: -31.1486 (%)

Earthquake Response - Dissipated Energy





Dissipated Energy from Earthquake= 24.14 N·mm

Dissipated Energy from Cyclic Displacement= 2884160127.16 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 0.000 [%]

ZONE 1: VERY MINOR DAMAGE

